ServiceOps complete platform manual

Version 1.27.22 · Visual walkthrough edition

Contents

See `docs/MASTER_REFERENCE.md` for a single merged document that also includes the governed backlog, traceability matrix, ITIL hierarchy reference, BP-001 blueprint traceability, REST API reference, and a fresh whole-application audit alongside this manual — useful as one-stop context for an AI agent or new engineer.

1. Purpose and operating model

ServiceOps is an enterprise service-management platform for requesters, fulfillers, team managers, CCB members, and platform administrators. It includes incident, request, problem, change, catalog, knowledge, CMDB, asset, SLA, approval, audit, analytics, and enterprise workspaces.

No deployment mechanism can eliminate every infrastructure or operator failure. The ServiceOps production standard therefore uses prevention, validation, observability, tested recovery, least privilege, immutable releases, and documented rollback rather than claiming infallibility.

2. Roles and teams

Role	Normal responsibilities
Requester	Submit and track requests and incidents, search knowledge
Agent	Triage, assign, fulfill, document, and resolve operational work
Manager	Agent work plus team oversight and manager approvals
CCB member	Review planned changes, risk, evidence, schedule, and backout plan
Administrator	Identity, service operations settings, CMDB, audit, and platform operation

CoreApps, Database, Network, Windows, Unix, and SSD are standard fulfillment teams. Administrators can map AD groups such as `gg_unix` to the Unix team; membership reconciles whenever the user logs in through AD/LDAP. Team-manager appointment and CCB approval authority remain explicit administrative decisions. Only active users granted `CCB approver` authority participate in new CCB approval gates. Normal and emergency changes follow manager assessment and CCB authorization.

Operational ownership is separate from approval authority. Every incident, service request, and change has one owning IT fulfillment team. An active member of that team, its configured manager, or a platform administrator can change operational fields. Assignees must remain active members of the owning team. Active members and managers of every IT fulfillment team can read all incidents and changes through lists, search, direct URLs, dashboards, analytics, task boards, history and attachments. This shared visibility supports triage, handoffs and operational awareness, but only the owning team or a platform administrator can change the parent ticket. A named approver can see the record needed for the decision, but CCB membership grants approval authority only and does not grant cross-team operational control. Administrator actions remain audited.

3. End-user guide

REST API documentation

The complete integration guide is maintained in `docs/API_REFERENCE.md` and is served by the running application at `/api/v1/docs`. The machine-readable OpenAPI 3.1 contract is `/api/v1/openapi.json`. It documents authentication, scopes, tenant and team authorization, field projections, idempotency, pagination, ticket operations, workflow triggers, monitoring ingestion, errors, retries, cURL and Python examples, and the explicit current compatibility boundary.

Governed record projections

Record visibility and field visibility are separate controls. Central tenant-aware object policies decide whether a user can discover or access a record. The validated Git-backed policy in `config/field_projections.json` then decides which fields may leave the application for that resource and audience.

The registry covers every supported record REST response, signed audit export, JSON search result, monitoring acknowledgement, workflow acknowledgement and interactive mutation acknowledgement. Unknown resources, unknown audiences, duplicate fields and fields outside a resource allowlist fail application startup. Requester ticket projections exclude internal assignment details; internal projections are limited to agent, manager and administrator audiences. Adding a new record API or export requires a reviewed registry binding and adversarial projection test.

Browser request security

Every state-changing browser request requires a session-bound CSRF token. ServiceOps inserts tokens into all rendered POST forms and supplies the token to interactive workspace actions through a dedicated request header. Tokens rotate after authentication. Missing, expired or mismatched tokens fail closed with HTTP 400 and make no data change.

Session cookies are HttpOnly and SameSite=Lax, and `SESSION_COOKIE_SECURE` defaults to `true` everywhere (Docker Compose, Helm, and the RPM package), not only in a production Helm profile. Outside test mode, the application refuses to start with insecure cookies unless `ALLOW_INSECURE_SESSION_COOKIES=true` is also set explicitly; that escape hatch exists only for an isolated plain-HTTP localhost/development deployment and must never be set in production. `SESSION_LIFETIME_MINUTES` controls the bounded session lifetime.

A request whose authenticated user has no resolvable `tenant_id` is logged out and rejected with HTTP 403 rather than silently defaulted to a tenant -- tenant resolution fails closed.

A Content-Security-Policy header (`default-src 'self'`, restricting script, style, image, font, and connection sources to the same origin) is sent on every response.

Database migration gate

Production schema changes are controlled by Alembic revisions. Compose runs the migration gate before initialization. Kubernetes uses a dedicated Helm pre-install/pre-upgrade migration Job; normal application replicas set `AUTO_MIGRATE=false` and refuse to start unless the database is already at the required revision.

Revision `20260726_0001` adopts a complete existing ServiceOps schema without rewriting operational records, or creates the schema for a fresh installation. Its downgrade is intentionally non-destructive: removal of an adopted baseline requires restoring the validated pre-migration backup. Every later schema revision must provide and test its own one-revision downgrade.

Revision `20260726_0002` creates the installation's default tenant and assigns all existing tenant-owned root records to it without deleting or recreating operational data. The application constrains ticket, request, enterprise, knowledge, asset, catalog, CMDB, notification, administration, approval, and global-search roots to the signed-in user's tenant. Cross-tenant direct identifiers return HTTP 404. Automated migration evidence covers fresh installation, existing-schema adoption, one-revision downgrade, roll-forward, row preservation, and tenant backfill. A deployment still requires a validated backup and an environment-specific rehearsal before production promotion.

For bundled PostgreSQL, run the guarded rehearsal before promoting an upgrade:

```
./serviceops rehearse-migrations 100000
```

The command clones the installed database into a database whose name must end in `_migration_rehearsal`, adds the requested number of isolated representative records, performs a one-revision downgrade and roll-forward, and compares table counts plus stable identity/content fingerprints. A cleanup trap removes the clone on success or failure. The command refuses external-database mode because those rehearsals must use an isolated database supplied by that provider.

Revision `20260726_0003` extends every audit event with a UUID, request correlation identifier, source context, integrity algorithm, previous-event hash, and event hash. Existing events are deterministically sealed into a tenant-specific legacy SHA-256 chain; new events use HMAC-SHA-256 with the deployment integrity key.

PostgreSQL and SQLite triggers reject audit UPDATE and DELETE operations at the database boundary. The administrator audit page verifies the entire tenant chain before displaying its status. Audit export is blocked if verification fails and otherwise returns ordered JSON evidence with a detached signature in `X-ServiceOps-Audit-Signature`.

Revision `20260726_0011` gives every HMAC event a signing-key identifier. Administrators can rotate forward from the audit page only after full-chain verification succeeds. Historical secrets remain encrypted for verification; neither events nor hashes are rewritten. The export response identifies its signing key in `X-ServiceOps-Audit-Key-ID`. `AUDIT_INTEGRITY_KEY_FILE` supports an initially mounted secret and takes precedence over the environment value.

The same page governs a minimum seven-year retention period, legal hold, and the requirement for external immutable evidence. Primary audit rows are never automatically purged. When audit streaming is enabled, each new audit event is committed transactionally to the durable outbox. Only active `siem` connections receive `audit.created` events; ordinary webhooks and Teams connections cannot receive this security stream. Delivery uses HTTPS, event identifiers, timestamps and HMAC signatures, with bounded retry and delivery evidence. The organization must still validate its actual immutable SIEM/WORM destination and retention controls.

Revision `20260904_0085` signs a bounded security context into each new audit event: source IP, optional forward-confirmed hostname, browser/OS label, user agent, language, authentication provider, HTTP method/path, request and trace IDs, sanitized referrer without its query string, and a one-way session reference. The audit page renders this evidence and verified exports include it. Cookies, authorization/CSRF headers, request bodies, proxy header chains, passwords, and tokens are deliberately excluded.

Protect `AUDIT_INTEGRITY_KEY` as a rotated secret independent from database operator access. When it is absent, ServiceOps uses the settings encryption key and finally the application secret. Changing the effective key without a governed chain rollover makes prior HMAC events unverifiable. Database append-only enforcement does not replace external immutable retention; export signed evidence to the organization's governed WORM or SIEM destination.

Declarative action and field authorization

`config/authorization.json` is the Git-controlled source for the initial role/action vocabulary. It declares discover, read, public/internal comment, create, update, assign, accept, transition, resolve, close, reopen, approve, delegate, relate, export, report, delete, purge, configure, administer, and security-administration actions. Startup fails if the policy is malformed or a role contains an unknown action. Browser handlers and future REST handlers use the same `serviceops_core.security` interface.

Object visibility remains relationship-aware and tenant-aware. Field projection is a separate decision: requesters viewing their own incident receive its public description, public comments, public attachments, state, priority, category, assignment summary, and ownership summary. They do not receive internal ticket history, work notes, change risk/plans, approval records, SLA internals, major incident coordination, affected-CI internals, operational tasks, or checklists. Active fulfillers retain the internal projection according to their object and team authorization. Client-supplied fields never expand that projection.

REST API v1

Administrators create and revoke REST identities under **Administration → API clients**. Each client is bound to one active user in the same tenant and inherits that user's object, relationship, field, and action authorization. Explicit scopes further restrict the client. Plaintext bearer tokens are shown only in the creation response; ServiceOps stores an HMAC hash and prefix, never the recoverable token. Protect the independently generated `API_TOKEN_PEPPER`; changing it revokes every existing token by design.

The initial `/api/v1` contract provides access-aware cursor-paginated ticket listing, individual ticket retrieval, incident creation, controlled ticket updates, and an OpenAPI 3.1 discovery document. Unsafe operations require an `Idempotency-Key`. Repeating identical content replays the stored JSON result; reusing a key with different content fails with HTTP 409. API errors include the correlated request identifier. Writes enter the append-only audit chain with the acting user and API client identifier.

Responsive PWA

ServiceOps publishes a dynamic company/instance manifest and root-scoped service worker. The worker caches only CSS and JavaScript shell assets. It does not intercept or cache HTML, authentication, API responses, tickets, requests, attachments, or other operational records. Installation outside localhost requires HTTPS. Offline operational records remain explicitly deferred until encrypted device storage, remote revocation, and data-loss policy are approved.

Durable integrations and monitoring

Revision `20260726_0005` adds a PostgreSQL-backed outbox, per-channel delivery evidence, encrypted webhook/Teams connections, authenticated monitoring sources, and deduplicated monitoring events. Application transactions create notifications and outbox events together. The separate Compose/Kubernetes worker claims due events with `FOR UPDATE SKIP LOCKED`, delivers them, records each attempt, retries with bounded exponential delay, and moves an event to `Dead` after five failed processing attempts. A successful channel is not sent again during retries of another channel.

SMTP is configured post-installation under platform settings. STARTTLS is enabled by default; hostname, port, account, encrypted password, and from address remain administrator-controlled. Signed webhooks use HTTPS and carry event ID, timestamp, and `HMAC-SHA-256` signature headers. Teams connections use the same durable worker with Teams-compatible message payloads. Literal loopback, private, link-local, and non-HTTPS webhook targets are rejected at configuration time. At delivery time the application re-resolves the destination hostname and rejects it if it now resolves to a non-global address, disables automatic redirect-following, and manually re-validates and re-resolves the target on every redirect hop (up to three), closing most of the DNS-rebinding/redirect-SSRF gap without relying solely on network egress controls. A narrow TOCTOU window remains between that re-resolution and the HTTP client's own connection-time DNS lookup; true IP pinning or a controlled outbound proxy is tracked as future work. Defense-in-depth egress firewall/DNS controls at the network layer are still recommended in production.

Administrators create monitoring sources under **Integrations** and bind each source to an active IT fulfillment team. The source token is displayed once and stored only as a hash. `POST /api/v1/monitoring/{source_id}/events` requires that bearer token, validates severity and payload limits, deduplicates by source/external ID, creates an EVT record, assigns an EVTASK investigation to the configured team, maps critical/high/medium/low severity to P1/P2/P3/P4, and records ingestion in the append-only audit chain.

Sign in

The sign-in page offers the identity methods enabled by the administrator: Keycloak/OIDC (a single **Continue with Keycloak** button), a local administrator account, and/or a corporate AD/LDAP directory account — when both local and directory sign-in are enabled, a dropdown lets the user pick which one they're using. For directory accounts, the username field accepts any of the three standard Windows login forms — the bare username (`jsmith`), the UPN form (`jsmith@company.com`), or the down-level logon form (`CORP\jsmith`). The field's own ghost text shows the site's actual domain as an example (e.g. `jsmith or jsmith@corp.example.com`), derived automatically from the configured LDAP base DN rather than a generic placeholder — and switches to a plain "Username" placeholder when the dropdown is set to the local administrator account instead.

Workflow state integrity

Approval-derived states are controlled by the approval engine and cannot be selected manually. A change remains `Awaiting Approval` until its active manager and CCB gates complete; only then can it move from `Approved` to `In Progress`. The ticket form and visual task board use the same server-side transition policy. Administrators cannot cast another named approver's vote. After approval, the owning team or a platform administrator can start, resolve, reopen, or close the work. A member of SSD can read a Unix-owned ticket for operational awareness but cannot operate it merely because that person is a manager or CCB approver.

Enterprise records with requested approvals remain locked in `Awaiting Approval`. Catalog fulfillment tasks cannot begin until their RITM approval chain completes, and terminal records cannot be reopened through a generic update endpoint. Invalid transitions return HTTP 409 and make no data change.

ITIL related-record model

ServiceOps treats operational work as a governed network rather than a ticket chain:

Record	Purpose and supported relationships
INC	Restore service; parent/child INC, primary and affected CIs, PRB, fix CHG, caused-by CHG, converted REQ
PRB	Root cause, workaround, known error and permanent fix; many INCs, PTASKs, multiple CHGs and knowledge
PTASK	Independently assigned investigation or resolution work package under a PRB
CHG	Risk, authorization, schedule and overall implementation control; related INCs, PRBs, RITMs, CIs and services
CTASK	Planning, implementation, testing or review work assigned to a specific team under a CHG
REQ	User order container containing one or more RITMs
RITM	Catalog item, variables, approvals and fulfillment stage; may link to a controlled CHG
SCTASK	Independently assigned fulfillment work under a RITM, in parallel or after a predecessor

CTASK, PTASK and SCTASK are distinct records. A required open CTASK prevents its parent CHG from completing. A required open PTASK prevents its PRB from completing. A RITM completes only after all SCTASK work reaches a terminal result, and a REQ completes only after all its RITMs complete.

Major-incident coordination remains an extension of an INC. It records proposed or accepted major status, business impact, coordinator and communications; it does not create a separate incident prefix. If the administrator enables parent incident state synchronization, supported state changes propagate to linked child INCs and are recorded in each child history.

Ticket history and approval-safe change revisions

Every operational record page contains its own chronological history. The history records the actor, precise time, event, field, previous value, new value and supporting details for record creation, state/priority/assignment changes, comments, attachments, checklist actions, related records, CI links, PTASK/CTASK/SCTASK work, and approval decisions.

For a CHG, the following are material approval inputs: purpose and description, change type, risk, impact, implementation plan, test plan, backout plan, planned window and primary CI. Editing any of them:

1. preserves the previous chain and votes as historical evidence;
2. marks the previous approval chain `Superseded`;
3. increments the CHG plan revision;
4. returns the CHG to `Awaiting Approval`;
5. creates a new manager/CCB approval chain; and
6. sends an in-application reapproval notification to every configured

approver.

An old approval is therefore never silently applied to a materially different change plan.

Disposable full-function test fixture

`tools/load_test_fixture.py` is an explicit non-production utility and is never executed during normal startup. It must be used only after an intentional database reset. The fixture displays a permanent red warning banner and creates an administrator plus one manager/CCB approver for every IT team:

Team	Username	Password
Administration	`admin`	`admin`
CoreApps	`coreapps`	`coreapps`
Database	`db`	`db`
Network	`network`	`network`
Windows	`windows`	`windows`
Unix	`unix`	`unix`
SSD	`ssd`	`ssd`

It also creates disposable catalog items, a configuration item, an asset, a knowledge article, a service offering, AD mappings, team-manager assignments, Service Desk memberships, and CCB approval authority. Never expose an instance containing this fixture to a network or reuse it for production data.

`tools/load_demo_dataset.py` (added in 1.27.18) is a second, non-production loader with the same `--confirm-non-production` requirement and identical production-image exclusion. Instead of a minimal fixture, it builds a realistic, deep dataset for manual exploration and testing: ~19 configuration items across three business-service dependency trees (application → database → server, plus shared network and storage CIs) with ~32 CI relationships, two named users per IT team (manager + agent; managers are also added as CCB approvers), five plain requester accounts, and a representative spread of 8 incidents, 3 problems (each with problem tasks), 3 changes (Standard/Normal/Emergency, each with change tasks), 2 catalog requests, and 5 knowledge articles. Every seeded user's password equals its username. Seeded changes and problems set their `state` directly rather than driving the live approval engine, so they render immediately without a pending approval blocking the screen — submit a *new* change or request through the UI against this data to exercise the live approval workflow itself. Keycloak SSO, AD/LDAP, and local administrator. The local administrator is a break-glass account and should not be used for routine work.

Navigation

The compact left application navigator is a persistent accordion with Home, My work, Self-service, Management, Operations, and Administration applications. Opening one application closes the previous top-level application, while the application containing the current page opens automatically and the exact module stays highlighted. It remains narrow and never overlays record content. The top bar contains the sidebar toggle, global search, saved/recent utility icons, notifications, help, and preferences.

Use **Find menu** at the top of the sidebar to filter every module available to your acting role. Matching applications and nested groups open automatically; clearing the field restores the accordion. This only filters navigation. Use the top-bar global search to find records and navigation destinations together.

Administration expands without hiding the other applications. It contains Administration home plus a nested **User management** group for Users, Groups/teams/access, Roles/permissions, and Active sessions. Governance, platform settings, integrations, automation, API access, audit, system health, and platform tenants (superadministrators only) have one direct canonical link.

Menu destinations are canonical: a page appears in one appropriate sidebar or Administration-home location rather than under multiple labels. Acronyms use their product spelling consistently, including **CMDB** and **RT**. The main search bar searches both navigation destinations and tenant-visible records, including tickets, requests, requested

items, operational and catalog tasks, knowledge, enterprise work, CMDB configuration items, assets, and catalog items. Administrators can additionally find users, groups, and integration connections. Results preserve the same tenant, role, team, and record-access rules as their source pages.

Long administration workspaces use a sticky section navigator. Selecting a section scrolls it into view; continued page scrolling automatically updates the highlighted section, its `aria-current` state, and the URL fragment. The navigator itself remains stationary: it does not auto-scroll or reorder as the page moves. A clicked destination remains selected throughout smooth scrolling instead of flashing through intermediate sections. Every Administration child workspace begins with a persistent breadcrumb and a clear **Back to Administration home** action so administrators never have to infer a return path from the global sidebar.

The sticky Administration context bar is the only breadcrumb pattern on administration pages. Page headers do not repeat an uppercase Administration home path, and list toolbars do not add another return button. Nested pages use the same bar with an intermediate parent, for example **Administration home / Users and access / New user** or **Administration home / Integrations and delivery / RT import**.

Service-delivery administration follows page order exactly: Catalog and routing; Directory mappings; Team aliases; Directory synchronization; Team ownership; Governance groups; Approval authority; Freeze windows; Service offerings; Calendars and commitments. Governance groups are separated from service offerings so every navigation item has one non-overlapping scroll target and related people, change, and service controls stay together.

Change approval policy is owned by **Service delivery and governance → Change approval policy**, beside approval authority and freeze windows. Platform settings does not duplicate it. The final **Runtime environment** section is a read-only deployment summary, not another settings form. It reports the active deployment profile, database endpoint, upload storage, replica count, and ingress/TLS ownership, together with whether each value is changed through Docker Compose, Helm, a secret, a volume, or an ingress controller. Explicit fallback text is shown when a deployment value is not declared.

Ticket defaults are owned by **Service delivery and governance → Ticket defaults**. This includes the default priority used when a new ticket does not declare one and the option to synchronize parent incident state changes to child incidents. Both controls apply live, are audited, and are not duplicated under Platform settings.

The Platform settings introduction reflects its current contents: organization identity and branding, sign-in and directory providers, security limits, workspace defaults, email delivery, NetBox and RT connections, and the read-only runtime environment. It directs ticket, team, change, service, and SLA policy to Service delivery and governance. The categorized Preferences workspace controls density, font scale, high contrast, reduced motion, accessible tooltips, keyboard shortcuts, date/list presentation, whether the sidebar stays pinned open, and the start page. The user name in the navigation footer opens the self-service profile, where a user can maintain their name, email, title, phone, time zone and date format. Role, active state, department, team membership, manager authority and CCB authority are not self-service fields. ServiceOps is light-only; there is no theme selector (ADR-012).

The sidebar profile card is the single place a signed-in user confirms who they are and signs out. It shows the user's name and their currently active role directly beneath their avatar. A user who holds more than one granted role (see **Roles and teams**, §2) additionally sees an **Acting as {Role}** control immediately below their name — opening it lists every role they hold, marks the currently active one, and lets them switch which role authorization checks use for the rest of the session with one click, without signing out. A single-role user never sees this control; it only appears when `current_user.granted_roles` has more than one entry. Below the profile identity (and the role switcher, when present) is a full-width, clearly labeled **Sign out** button — both the icon and the word are always visible, not an icon-only affordance a user has to guess at.

System Health

Administration home → System → System Health is the operator-facing view of application errors and request activity, distinct from the Audit log (§Audit log), which records user actions rather than technical events.

The page has two views: the application error table (backed by the `ApplicationLog` database record, survives container restarts) and a raw request-log-file viewer reading the same JSON lines the container writes to `LOG_DIR`. Both views share the same combinable, Splunk-style filter bar — severity level, logger name (contains), request path

(contains), request ID (exact match), and a from/to date-time range; the log-file viewer additionally filters by HTTP method and status code. Filters combine with AND semantics and the same filtered result set is what gets exported, so what an operator sees on screen is exactly what they download.

Each view has its own export action supporting four formats — CSV and plain text for spreadsheet tools, JSON and NDJSON for SIEM/log-shipper ingestion (NDJSON is the bulk-ingest shape Splunk and Elastic both prefer) — via `GET /admin/system-health/errors/export` and `GET /admin/system-health/logs/export` (`format=csv|json|ndjson|txt`, admin-only, each export itself is audited). Exports are capped at 10,000 rows per request and inherit the same filters currently applied on screen.

Whatever detail appears in the in-app log viewer and the `LOG_DIR` file also appears, byte-for-byte identically, in `docker logs`/`kubectl logs` output — the same `JsonLogFormatter`/`RedactingFilter` pipeline that writes the file also writes to stdout, so an operator without shell access to the log volume never sees less detail via the container's own log stream than the in-app viewer promises.

Administrators use Administration home for a capability-oriented entry point. While an administration page is open, the normal workspace menu becomes a pinned, filterable navigator grouped as Overview, Users and access, ITSM configuration, CMDB, Workflow and automation, Data management, Integrations, Security, User interface, Development, Deployment, and System. Modules link to implemented ServiceOps capabilities and specific configuration sections; the UI does not advertise unsupported ServiceNow tools. Users and roles provides tenant-scoped search and an editable user record. Only administrators with `security_administer` may alter role, active state or department. AD-sourced team membership continues to reconcile from configured directory mappings and is not replaced by profile editing.

Incidents

Create an incident with a concise title, observable symptoms, business impact, category, and urgency. Agents classify, prioritize, assign, comment, attach evidence, execute checklists, observe SLA timers, resolve, and close it.

The incident record uses a dense, two-column operational form. Its header keeps Update and permitted Resolve actions visible, while the body exposes the number, caller, contact type, state, category/subcategory, impact/urgency, calculated priority, service offering, primary configuration item, assignment group, assignee, notification preference, short description, and description. The immutable Event history follows the form directly and records field-level old/new values with actor and timestamp. Authorization remains server-enforced: the owning team controls operational mutations even though active IT teams can read incidents.

The same task-derived interaction model is used for changes, problems and other enterprise operational records, and request containers. Every supported task record keeps its number/type identity and permitted actions in a consistent header, presents common state/priority/requester/assignment fields in the same two-column pattern, places field-level Event history immediately below the record, and then exposes type-specific governance, tasks, approvals, SLAs, attachments, work notes and related records. List workspaces use a consistent New/search/filter bar, encoded-filter summary, record count, pagination, priority indicators and operational columns.

Requests and catalog

Select a catalog item, supply its variables and business justification, and submit. ServiceOps creates REQ, RITM, approvals, and SCTASK records. Follow the request page for approval and fulfillment status.

For an approval-required item, the first approval is assigned to the requested-for employee's active line manager recorded in ServiceOps, including a manager profile provisioned from AD/LDAP before that manager has ever logged in. The assignment is snapshotted when the RITM is submitted so a later directory change affects future requests without silently redirecting an approval already under audit. ServiceOps never substitutes an administrator: an absent, inactive, self-referential, or cross-tenant manager relationship fails closed before any REQ/RITM is created. The second stage requires an active Service Desk member.

Each catalog item has an administrator-controlled default fulfillment route under **Administration home → Service delivery and governance → Catalog and routing**. The generated initial SCTASK inherits that team. Laptop and Software catalog items are routed to Windows by default. Administrators can route any current or future catalog item to another active fulfillment team without changing application code. The same screen creates and edits catalog items,

including name, category, description, delivery target, approval requirement, active availability and fulfillment team. New items require an explicit active fulfillment team. Legacy items without an explicit route use the active Service Desk as a controlled fallback; ordering fails closed if neither an explicit team nor Service Desk is active. Inactive items cannot be ordered through either the UI or a direct endpoint.

Catalog request visibility is need-to-know. A REQ and its RITMs are visible only to the requested-by/requested-for users, members or managers of the configured fulfillment team, teams assigned an SCTASK, specifically named approval voters, and platform administrators. The same policy protects the request list, direct URL access, dashboard counts and global search. Membership in an unrelated team, such as Unix, does not expose a Windows-routed request. Administrators retain complete visibility for platform oversight and audit.

Changes

Every change must state type, affected CI, owning team, risk, impact, planned window, implementation plan, test plan, and executable backout plan. Review conflicts before approval. CCB approval is authorization, not a substitute for technical validation or membership in the owning implementation team.

A change can affect more than one Configuration Item. The New Change form captures a primary CI (used for risk scoring and conflict/freeze detection against overlapping changes) plus an optional, repeatable **Additional configuration items** picker — click **Add another configuration item** for each further CI the change touches. Every additional CI is linked at the moment the change is created and appears on the change's own **Affected CIs** section alongside any CIs added later from the record itself. Adding an affected CI to an already-approved change is a material change: it invalidates the current approval chain and starts a new approval cycle, the same as editing the implementation, test, or backout plan.

Knowledge, CMDB, assets, and boards

Search knowledge before opening duplicate work. Use the CMDB relationship view to understand service impact. Asset pages track accountable inventory. The visual task board changes the underlying ticket state and therefore remains audited and role controlled.

Agentless discovery (Administration → CMDB → Discovery) finds Configuration Items and their connections from switches and other SNMP-speaking devices, without installing anything on the target — an administrator configures a discovery target (a single host or a CIDR subnet) with its own SNMP version/port/community string, encrypted at rest the same way other per-integration secrets are. A target can be run on demand or on a schedule (the same in-process worker loop that handles SLA breaches and LDAP sync). Each run reads standard MIB-II/IF-MIB system and interface facts, the device's ARP table, and LLDP neighbor advertisements for anything that answers SNMP. Most consumer and office devices — phones, laptops, most routers, smart-home gear — don't run an SNMP agent at all, so a device that doesn't answer SNMP is still checked for basic reachability (a handful of common TCP ports, never a port scan) and, if alive, recorded as a bare result — IP address plus a best-effort reverse-DNS name where the network's own DNS/router setup supports it — rather than being silently invisible; on a typical network expect far more devices found this second way than the first.

A run never creates a CI by itself. It stages every device found as a reviewable candidate; the discovery list shows a "Review N devices" link whenever a target has pending results, opening a page listing each one (name, address, class, vendor, and whether it came from SNMP or a bare liveness hit) with a checkbox per row. From there an administrator adds the selected devices, adds every candidate in one action, or discards the whole batch without touching the CMDB — nothing lands in the CMDB without an explicit decision. Only at that point does reconciliation run: it creates a CI per chosen device (guessing a sensible class — Network Switch, Server, Network Appliance — from what it saw) and a "Connects to" relationship for every LLDP-discovered neighbor pair where both sides were selected together. Reconciliation is deliberately conservative toward manual edits: a CI an administrator already classified by hand keeps its name, class, and vendor exactly as set on a later re-import — only its last-seen technical detail is refreshed — and a device already fully profiled via SNMP is never downgraded to a bare entry by a later liveness-only hit on the same address. This is a distinct, complementary path to the existing self-registering host agent (`tools/cmdb_sync_agent.sh`, which a host runs on itself); discovery is for devices — switches, appliances — that can't run an agent of their own. Nothing is ever scanned beyond a target an administrator explicitly entered.

Topology map (`/cmdb/topology`, linked from the CMDB page) renders every CI and relationship visible to you as an interactive, draggable graph — manually created and SNMP-discovered CIs are colored distinctly, and hovering an edge shows the relationship type. No screenshot exists yet for this view in the visual walkthrough below; it renders identically to how it looks live.

Release 1.27.18 enriches the Configuration Item record beyond the original name/class/environment/status/IP/owner fields: each CI now also carries a description, lifecycle state (Planned/In Use/Maintenance/Retired/Disposed, distinct from operational status), business criticality (Critical/High/ Medium/Low), serial number, vendor, model, physical/logical location, cost center, discovery source (Manual/Discovery scan/Import/API), install and warranty-expiry dates, an owning support group (in addition to the individual owner), and a free-form JSON attributes bag for anything not covered by a named column. CI-to-CI relationships are now tenant-scoped in their own right (`ci_relationship.tenant_id`) rather than relying solely on their parent/child CI's tenant, and a given relationship type between the same two CIs can no longer be created twice. Administrators manage all of this from `/cmdb` and `/cmdb/<id>/edit`; the REST CMDB auto-registration endpoint (`PUT /api/v1/cmdb/configuration-items`, `cmdb:write` scope, see `API_REFERENCE.md` §9) has **not** yet been extended to accept these new fields — it remains upsert-by-name over the original five fields only, so richer CI data must currently be entered through the web UI. Migration: `20260729_0023_cmdb_enrichment.py`.

Administrators queue **CMDB → Import → Sync from NetBox** as a dry-run preview before applying reconciliation. The dedicated worker reads configurable bounded pages (100 records by default), permits only one tenant job at a time, persists progress and terminal evidence, and checks cancellation between API pages. The page displays live phase/count/progress and a safe Cancel control; closing the browser does not stop or lose the job. ServiceOps treats the NetBox inventory schema conservatively:

- device name, serial, device-type manufacturer/model, primary management IP,

site/location, and rack placement become structured CMDB fields;

- NetBox operational status maps separately from CMDB lifecycle (for example,

`offline` becomes **Down / In Use**, not Retired; `staged` becomes **Maintenance / Planned**);

- common functional role terms normalize to controlled classes such as Server,

Switch, Router, Firewall, PDU, Storage, Load Balancer, or Wireless Access Point; a custom role is retained as `NetBox: Role` and uses class Device;

- platform remains an operating-system attribute, and a VM cluster remains a

logical-grouping attribute. Neither is mislabeled as hardware Model or physical Location; a VM's site is used for Location when supplied;

- NetBox tenant is retained for reference and is never treated as the owning

ServiceOps support group. Operational ownership and business data remain governed in ServiceOps;

- typed identifiers (`dcim.device:<id>` and

`virtualization.virtualmachine:<id>`) prevent independent NetBox object tables with the same numeric primary key from overwriting one another;

- fields removed at the source are cleared on the next applied sync. Dry run,

per-record error isolation, component permission warnings, and separate physical-device/VM counts make the effect reviewable.

- current NetBox device and VM serializers are consumed without flattening

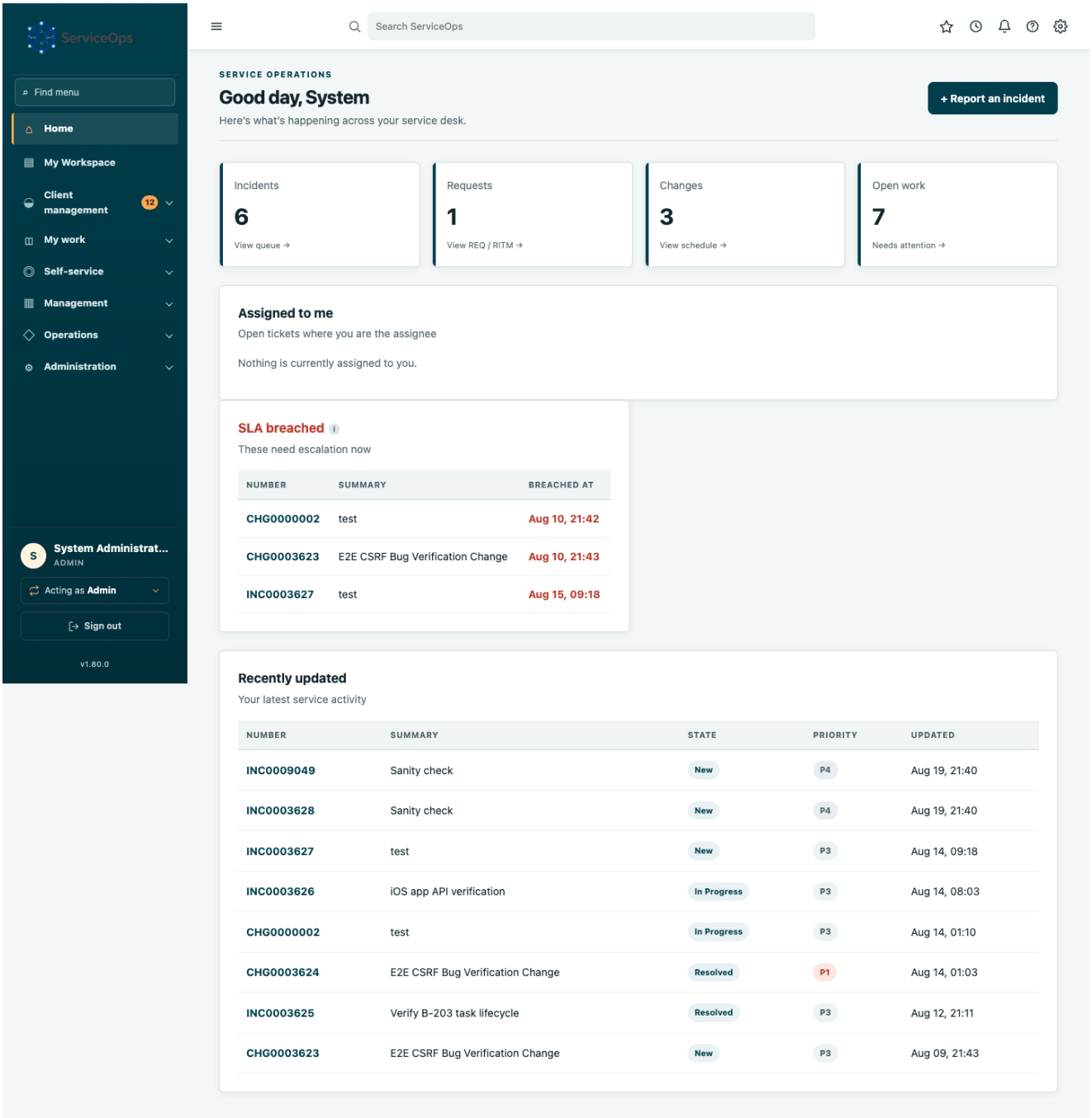
unlike concepts: assigned IP/DNS/VRF details, physical and virtual interfaces, VLAN/cable context, console/power/cooling components, modules, bays, inventory items, virtual disks, owner, coordinates, configuration context, timestamps, tags and custom fields are retained as namespaced attributes; rack location/group/facility/status/role/type/dimension/weight metadata is retained on the rack itself.

Use a read-only NetBox token. NetBox v2 tokens beginning `nbt_` are sent with Bearer authentication; legacy v1 tokens remain supported with Token authentication. TLS verification is enabled by default; add the internal CA certificate rather than enabling the explicitly marked insecure bypass.

3A. Visual walkthrough

This chapter shows every major workspace as it actually renders, captured directly from a running instance, so a new user or administrator can recognize each screen before they open it for the first time. Screenshots were captured against a local development instance seeded with realistic sample data (`tools/load_demo_dataset.py`); a production instance looks identical in layout and differs only in the records shown.

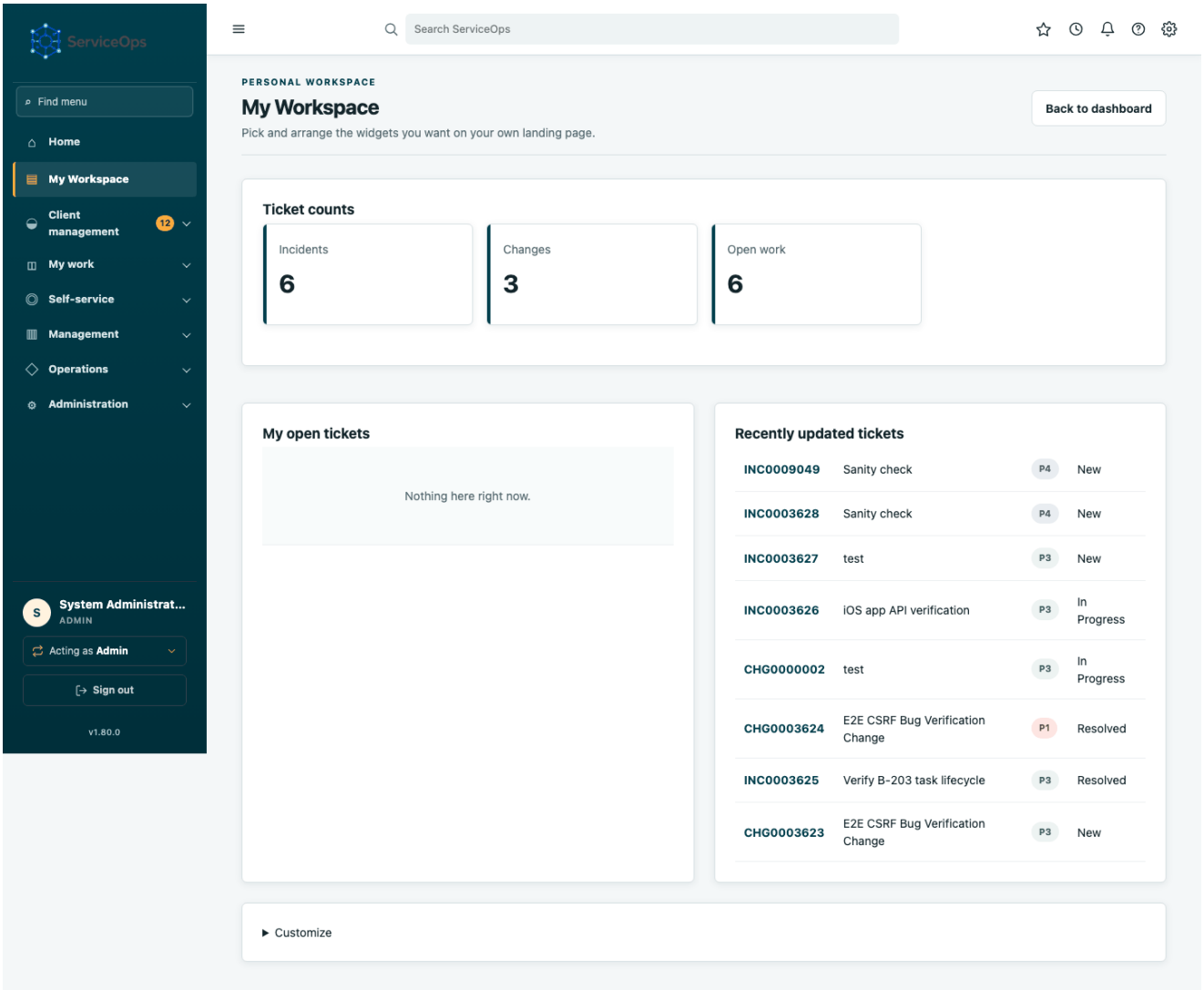
Dashboard



The ServiceOps dashboard: stat tiles for Incidents, Requests, Changes and Open work across the top, followed by Assigned to me, SLA breached, SLA at risk and Recently updated panels.

The dashboard is the landing page after sign-in. The top row of stat tiles gives an at-a-glance count of open Incidents, Requests, Changes, and total Open work; each tile is a link straight into the matching filtered list, and the Incidents tile turns red with a P1/P2 breakdown whenever a high-priority incident is open. Below that, **Assigned to me** lists your own open tickets ordered by priority. **SLA breached** and **SLA at risk** only appear when there is something in them — an empty service desk means an empty dashboard, not two panels permanently reading "nothing here." **Recently updated** shows the last eight records touched anywhere you have visibility into, useful for picking back up after a context switch. Administrators can turn any of these panels on or off tenant-wide from **Administration home → Platform settings → Workspace defaults**.

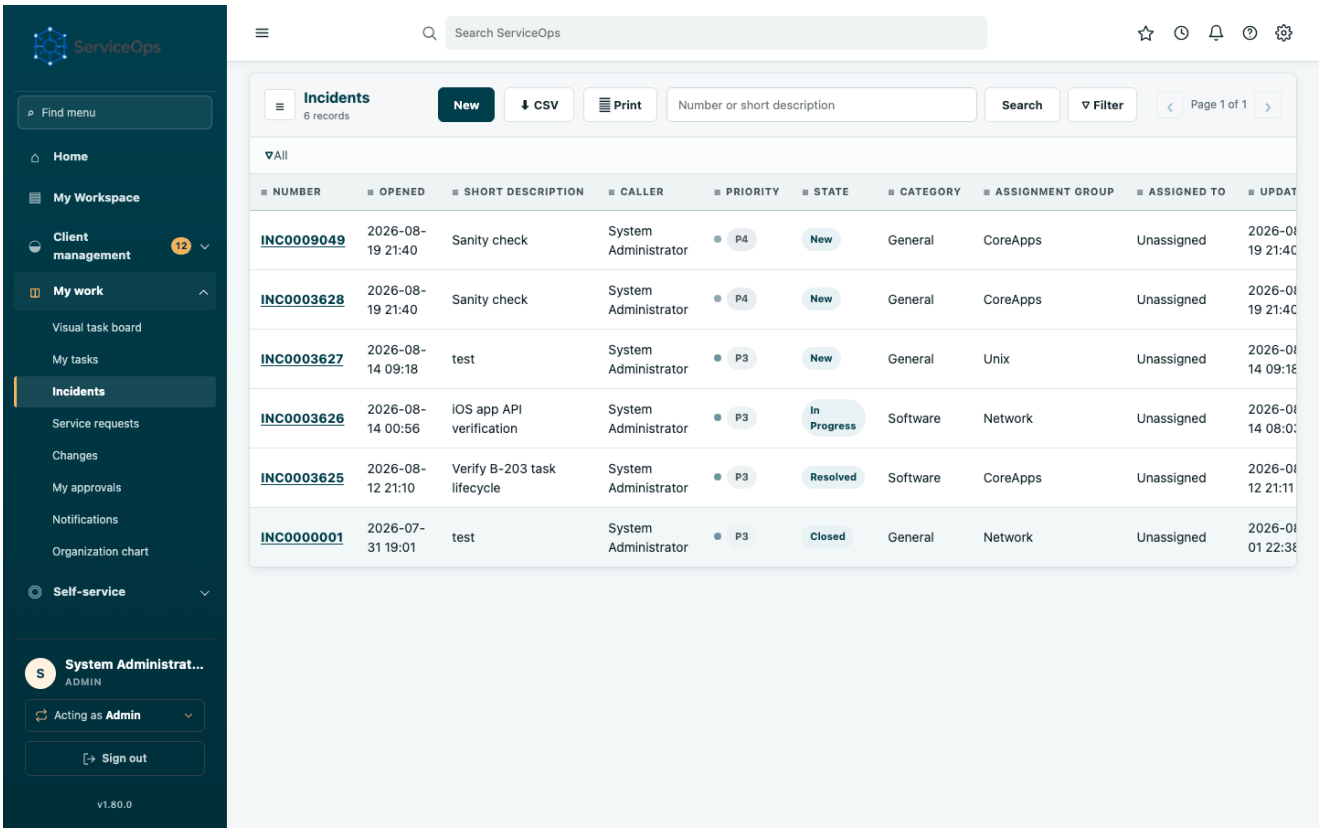
My Workspace



My Workspace: a per-user, drag-configurable landing page built from a closed catalog of widgets -- ticket stats, my open tickets, recent tickets, SLA-at-risk, approvals awaiting me, favorites, recently viewed, and notifications.

My Workspace (`/workspace`) is an alternative, personal landing page each user can lay out to their own preference, sitting alongside the shared Dashboard rather than replacing it. Its widget catalog is closed and code-defined — ticket stats, my open tickets, recent tickets, SLA-at-risk, approvals awaiting me, favorites, recently viewed, and notifications — not a general-purpose page designer; an administrator can disable individual widgets tenant-wide from Platform settings, and each user's own layout is saved independently of everyone else's.

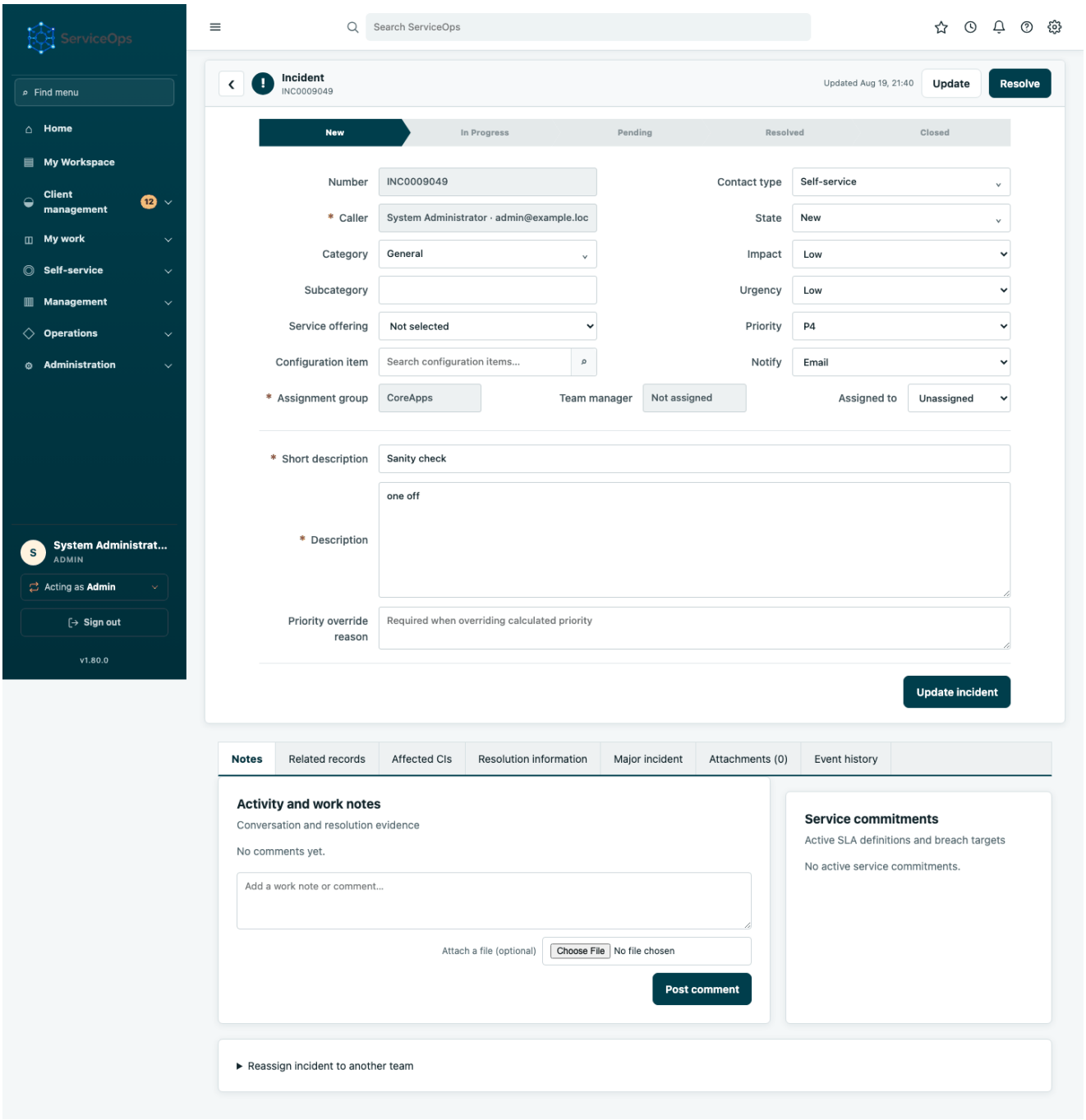
Incidents list



The Incidents list workspace: a searchable, filterable, paginated table with Number, Opened, Short description, Caller, Priority, State, Category, Assignment group, Assigned to and Updated columns.

Every ticket list in ServiceOps (Incidents, Changes, Requests) shares the same list-workspace layout: a toolbar with a **New** button, a state filter, a free-text search box that matches both the record number and its short description, and pagination controls showing the current page and total record count. The table itself is sorted by most-recently-updated first and color-codes priority with a small dot (red for P1, amber for P2) so urgent work is visually distinct while scrolling a long list. Clicking any row's number opens that record's detail page.

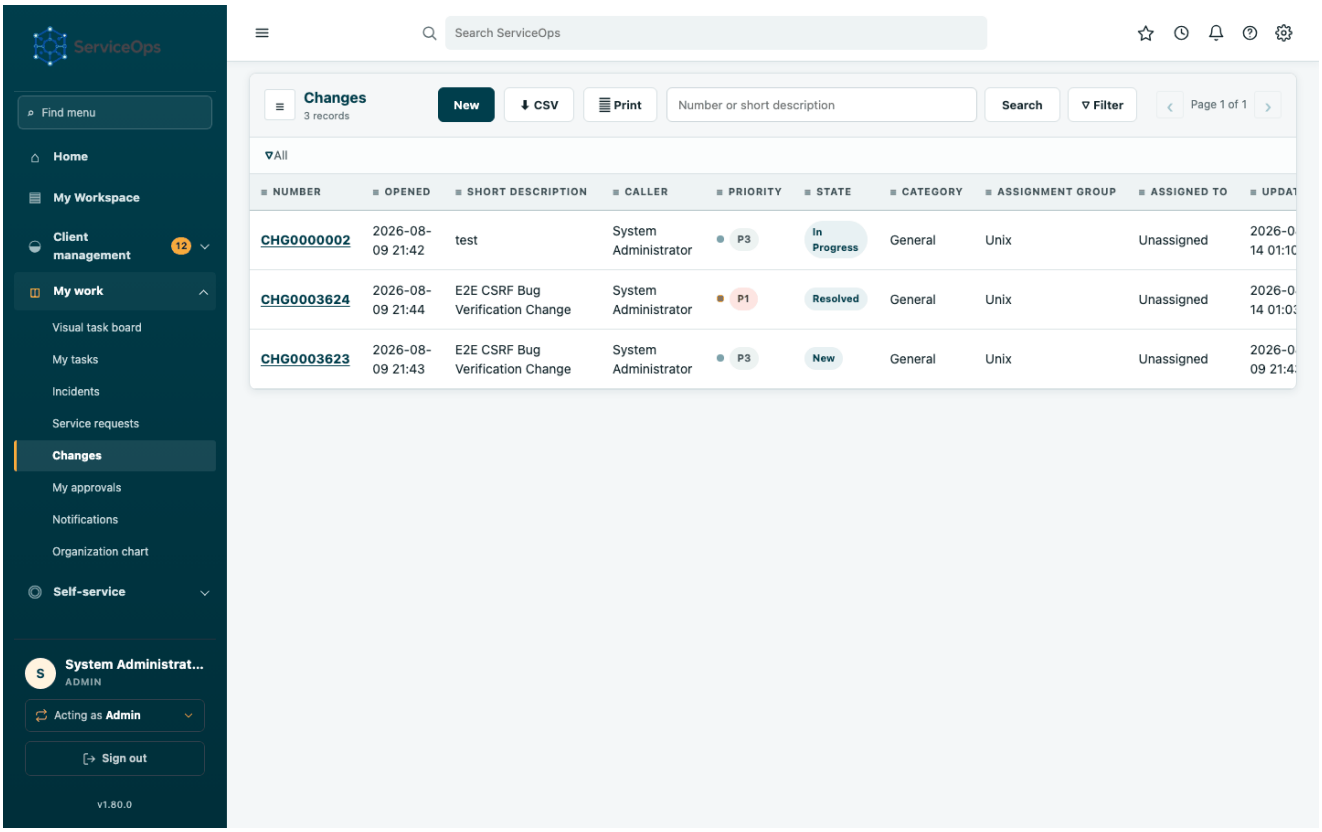
Incident detail



An open incident record: a lifecycle stepper (New → In Progress → Pending → Resolved → Closed) across the top, a two-column form with State, Priority, Category, Assignment group and Assigned to fields, and an Event history timeline below.

The incident detail page is built on the same "record shell" used for every ITIL record type in ServiceOps: a header bar with the record number and permitted actions (Update, Resolve), a horizontal lifecycle stepper showing exactly which state the record is in and which states are still ahead, a dense two-column form for the operational fields, and a chronological Event history immediately below recording every field change with who made it and when. Only an active member of the incident's owning fulfillment team (or an administrator) can change these fields — other teams can open this same page to read it for cross-team awareness, but the form controls stay disabled for them.

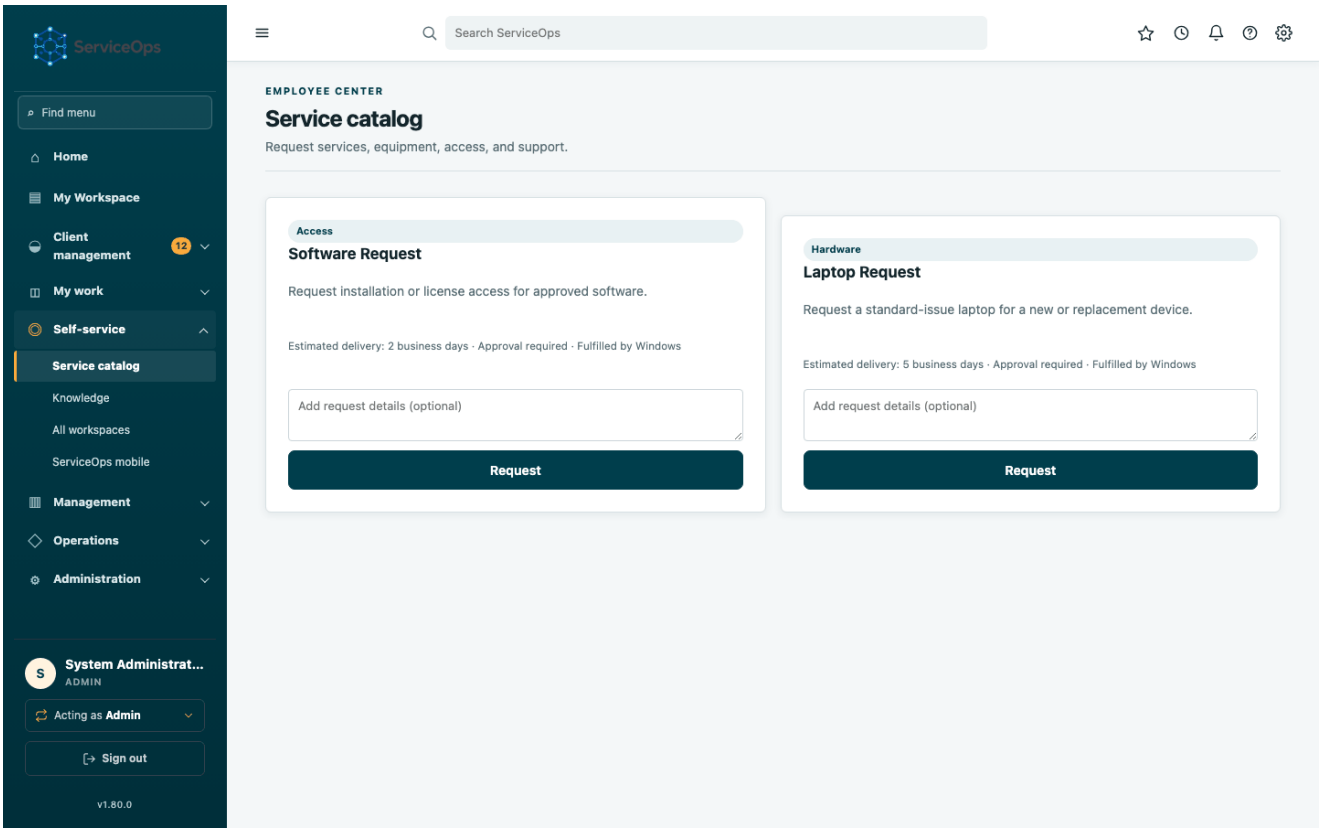
Changes list



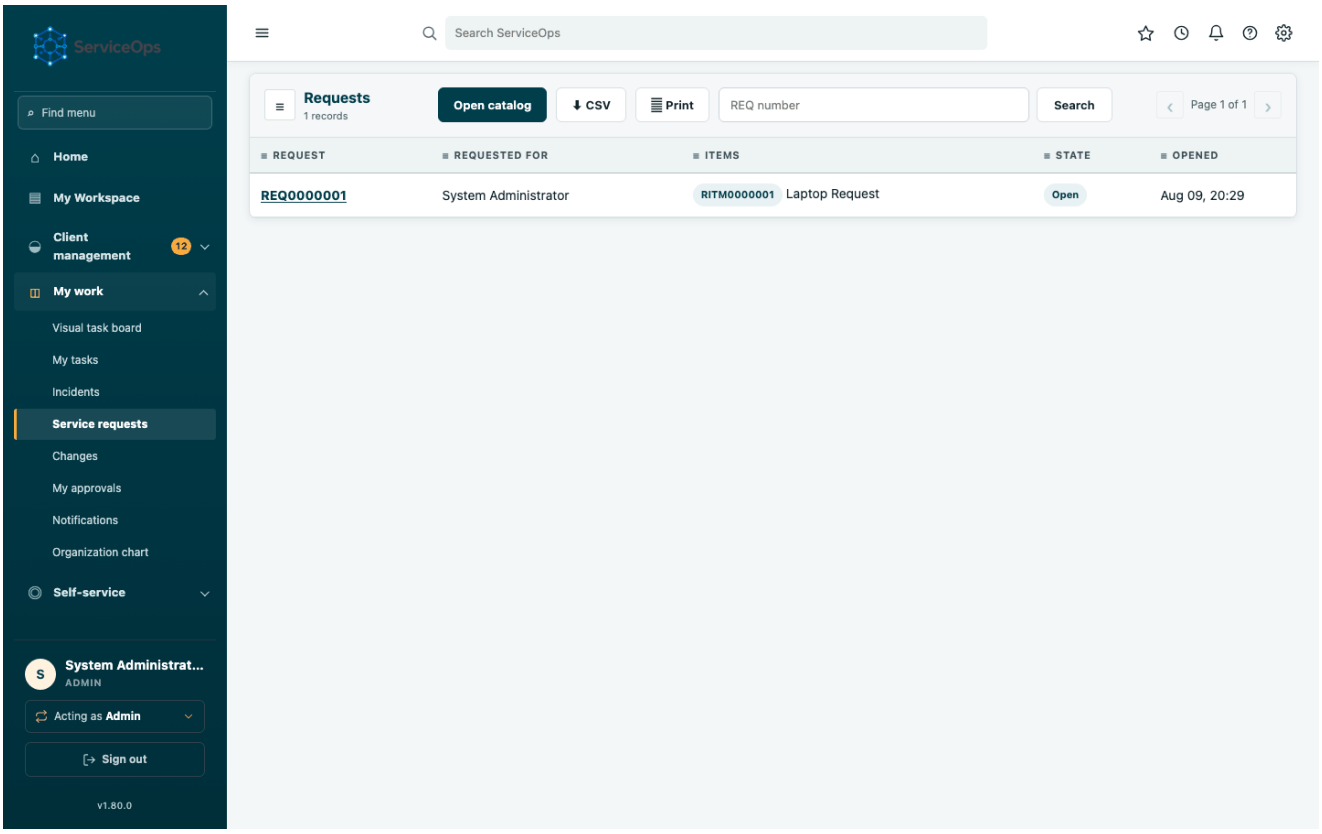
The Changes list workspace, filtered to a specific state, showing CHG-numbered records with their risk-bearing priority and assignment group columns.

Changes use the identical list layout as incidents, but every row here represents a controlled production modification rather than a service interruption. Look at the State column to distinguish a change that's still `Awaiting Approval` from one already `Approved` and scheduled, or already `Implemented`. Standard, Normal and Emergency changes are all shown together in this list — open the record itself to see which governance path applies (see "Changes" earlier in this manual for the approval rules each type follows).

Requests & catalog



The service catalog: a grid of orderable catalog items (Laptop Request, Software Request) each with a description, estimated delivery time, approval requirement and fulfilling team, and a "Request" button.

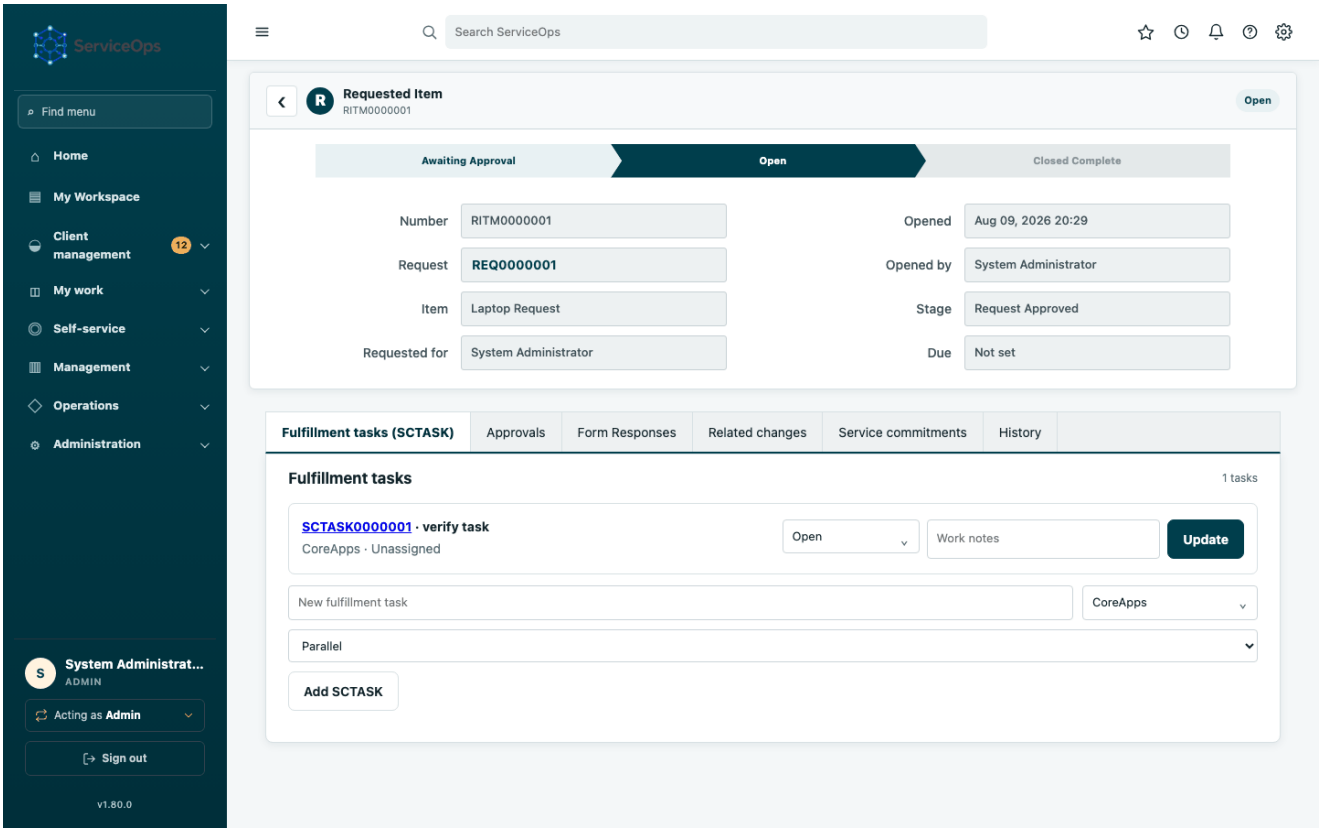


The Requests & RITMs list: REQ-numbered containers with Requested for, Items, State and Opened columns.

The **Service catalog** page is where end users browse and order predefined items. Each card states up front whether approval is required and which team will fulfill it, so there are no surprises about turnaround time. Submitting the form creates a REQ (the order) and one RITM per item ordered; the **Requests & RITMs** list is where you track

those containers afterward, independent of the catalog itself.

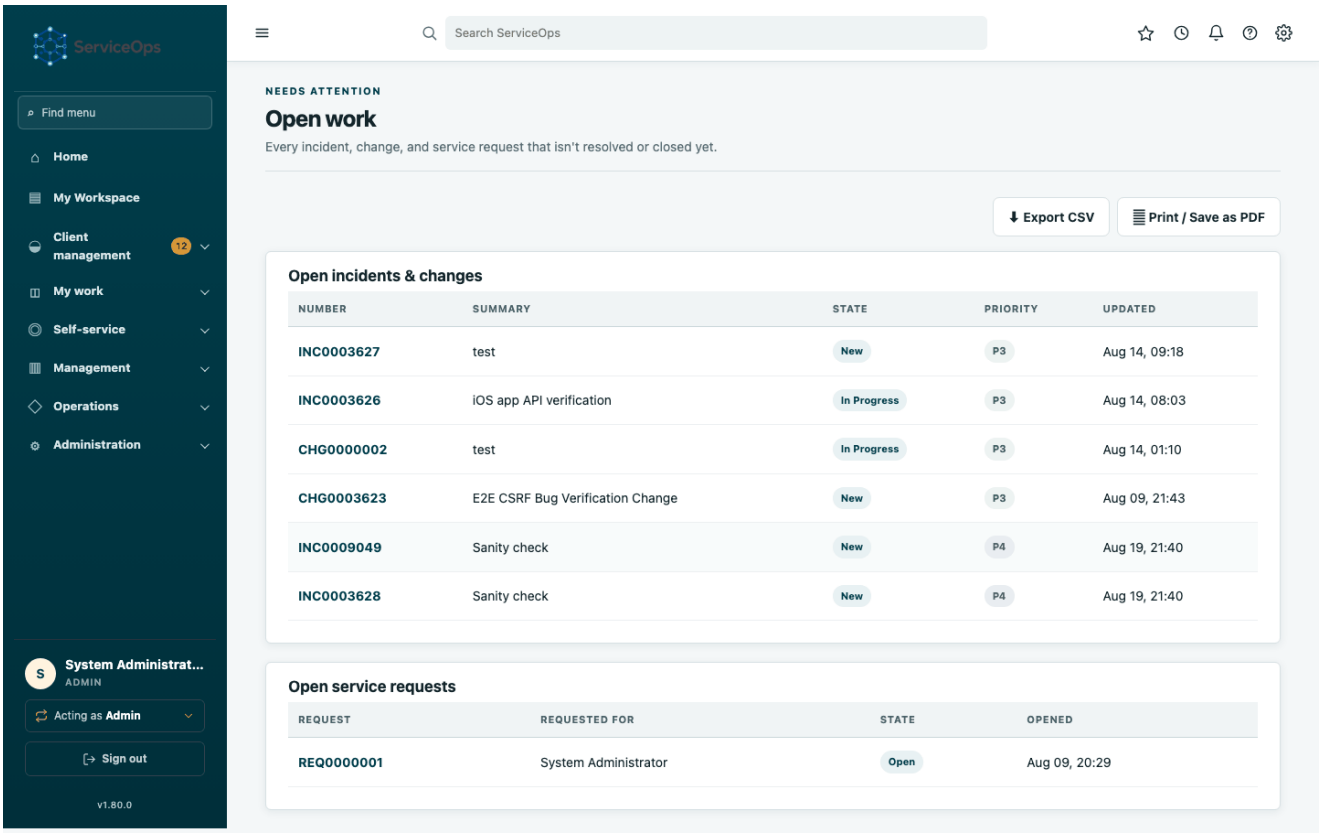
Requested Item (RITM) detail



A RITM detail page showing its lifecycle stepper (Awaiting Approval → Open → Closed Complete), Opened/Opened by fields, and a Fulfillment tasks (SCTASK) panel below.

Opening a REQ shows its constituent RITMs; opening a RITM shows this page. The lifecycle stepper here reflects the RITM's own three-stage journey — approval, open fulfillment, and completion — distinct from the parent REQ's own state. Below the header, the **Fulfillment tasks** panel lists every SCTASK generated to actually deliver the item, each independently assignable to a fulfillment team; the RITM itself only reaches `Closed Complete` once every one of its SCTASKs finishes.

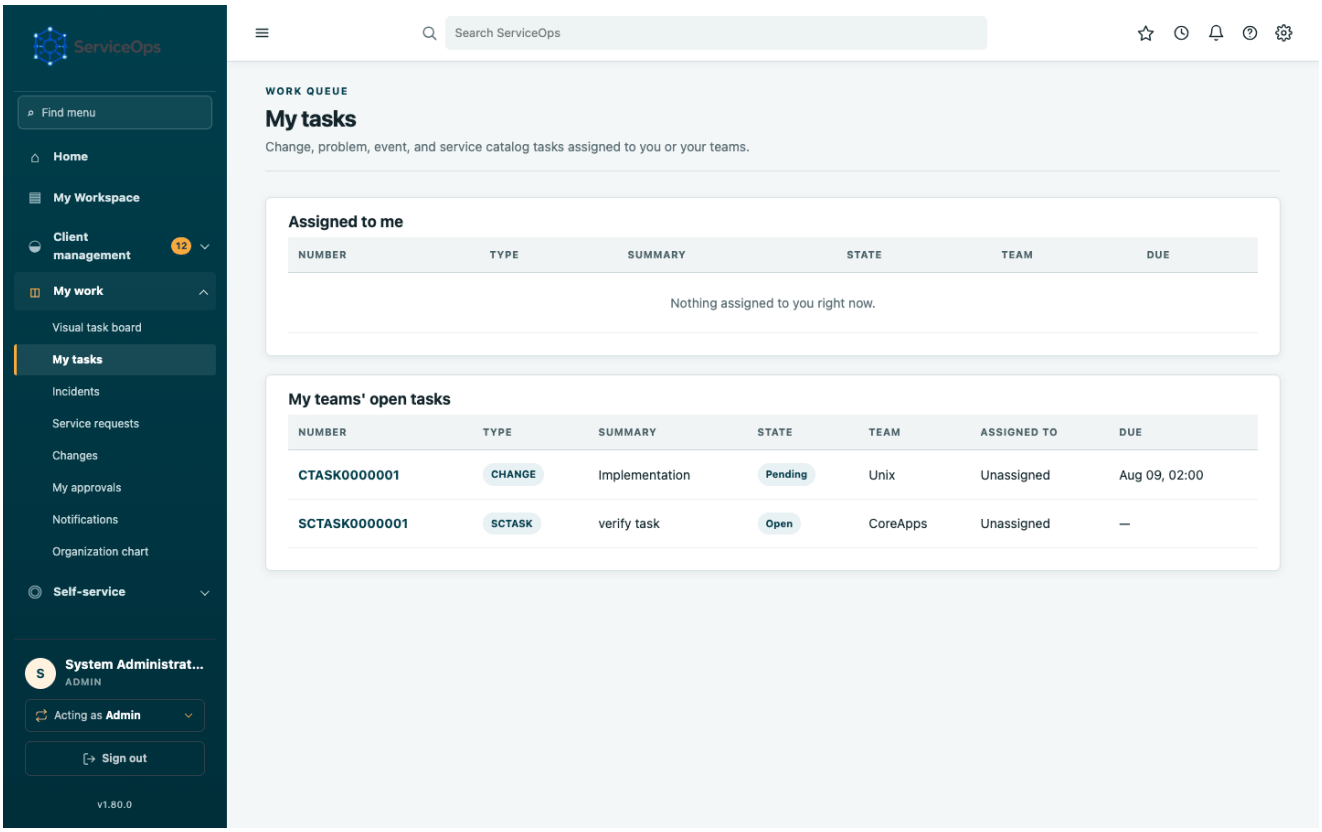
Open work



The Open work page: every incident, change and service request across the tenant that isn't resolved or closed yet, in one combined view.

This is the single cross-cutting view of everything still outstanding, combining incidents, changes and requests into one list rather than three separate ones. It is capped at 200 rows per section to stay responsive on a busy service desk — a truncation notice appears with links to the fully filterable Incidents/Changes lists if there's more to see. Use the priority filter (linked from the dashboard's P1/P2 alert) to narrow straight to the highest-urgency backlog.

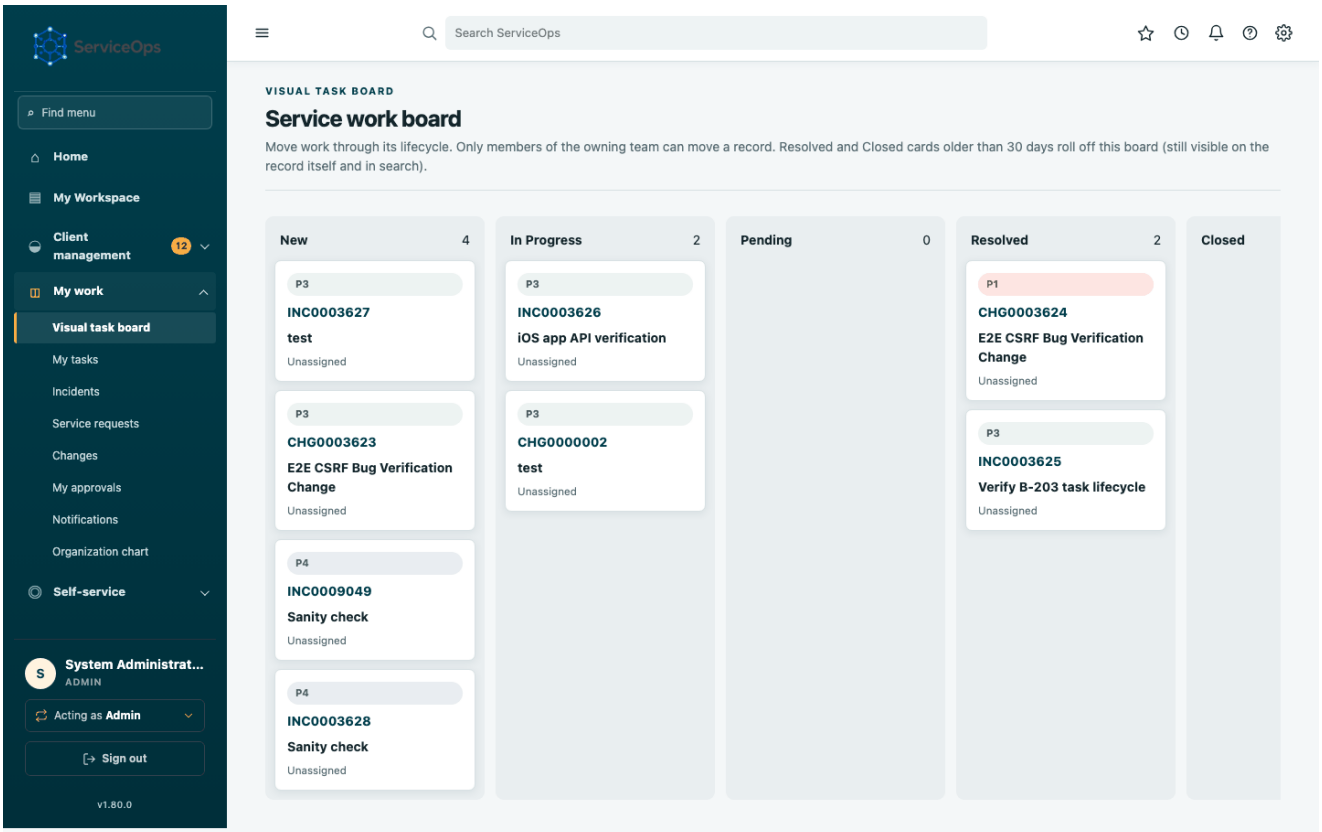
My tasks



The My tasks page: a queue of operational tasks and catalog fulfillment tasks assigned to the signed-in user, plus a secondary list of unassigned team work.

Agents and managers use this page as their personal work queue. **Assigned to me** is exactly that — CTASKs, PTASKs and SCTASKs with your user ID as the assignee, sorted so the nearest due date surfaces first. **Team tasks** below it shows everything else open against a team you belong to but not yet picked up by anyone, which is where you'd look to grab the next piece of unassigned work. The sidebar's "My tasks" link itself carries an amber badge whenever you have open work assigned, so you don't have to open the page to know something is waiting.

Visual task board



A Kanban-style visual task board with New, In Progress, Pending, Resolved and Closed lanes, each containing draggable ticket cards.

The task board gives a Kanban view of tickets you're permitted to manage. Dragging a card between lanes changes the underlying ticket's state exactly the same way editing the record's State field would — it is a different interaction model over the same governed transition rules, not a shortcut around them, so an invalid drag (e.g. skipping a required approval state) is rejected the same way an invalid form edit would be.

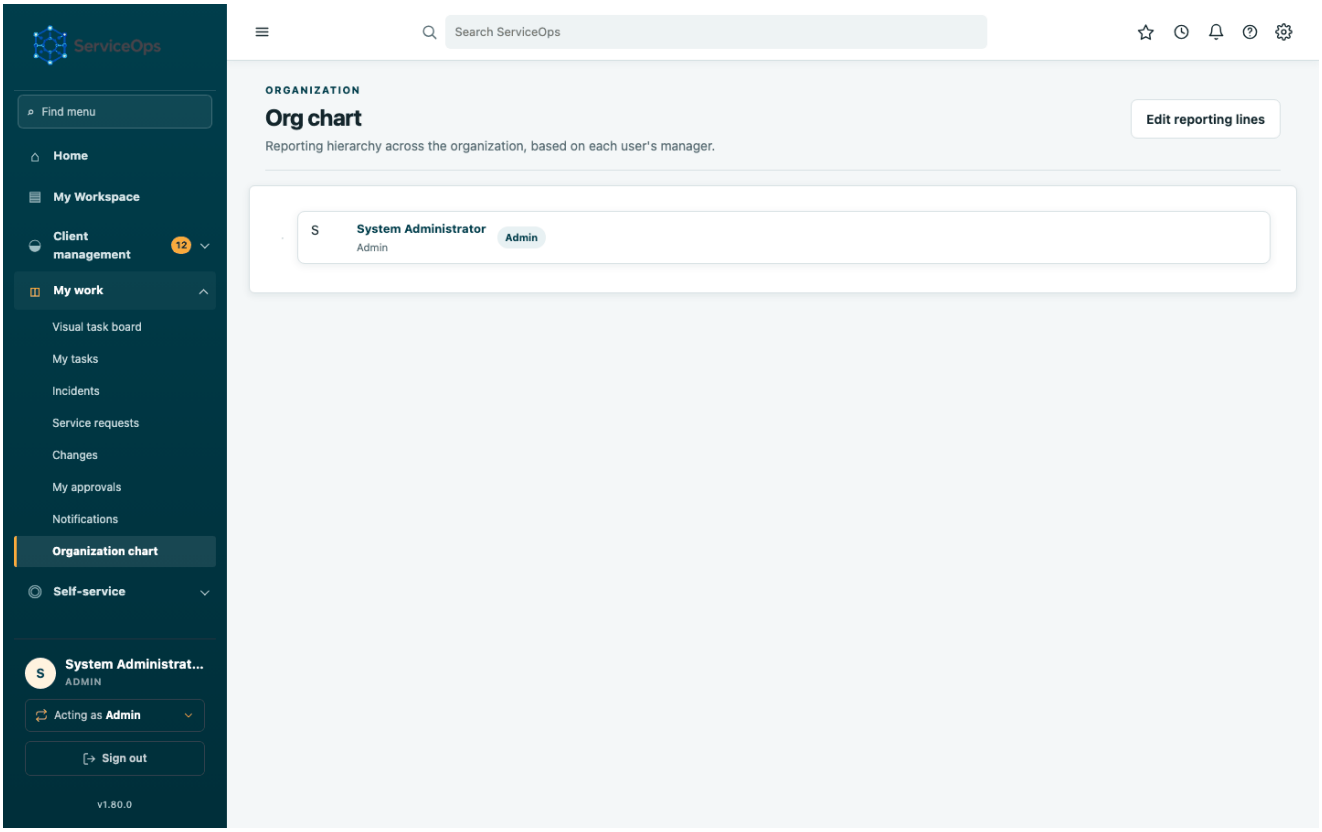
CMDB & service map

The CMDB page: a table of configuration items with Name, Class, Environment, Status, Lifecycle, Criticality, Location, Owning team and Owner columns, followed by a list of CI-to-CI relationships.

The Configuration Management Database is the system of record for every server, application and service ServiceOps tracks, and how they depend on one another. Each CI carries a lifecycle state (separate from its operational status —

a CI can be operationally `Operational` while its lifecycle state is `Planned` ahead of go-live, for example), a business-criticality rating used elsewhere in change-conflict detection, and an owning support team. Relationships beneath the CI table (e.g. "Customer Portal — Depends on → Payroll Database") are what powers change-conflict detection: scheduling a change against a CI that another in-flight change already touches, directly or through a relationship, surfaces a conflict warning before approval.

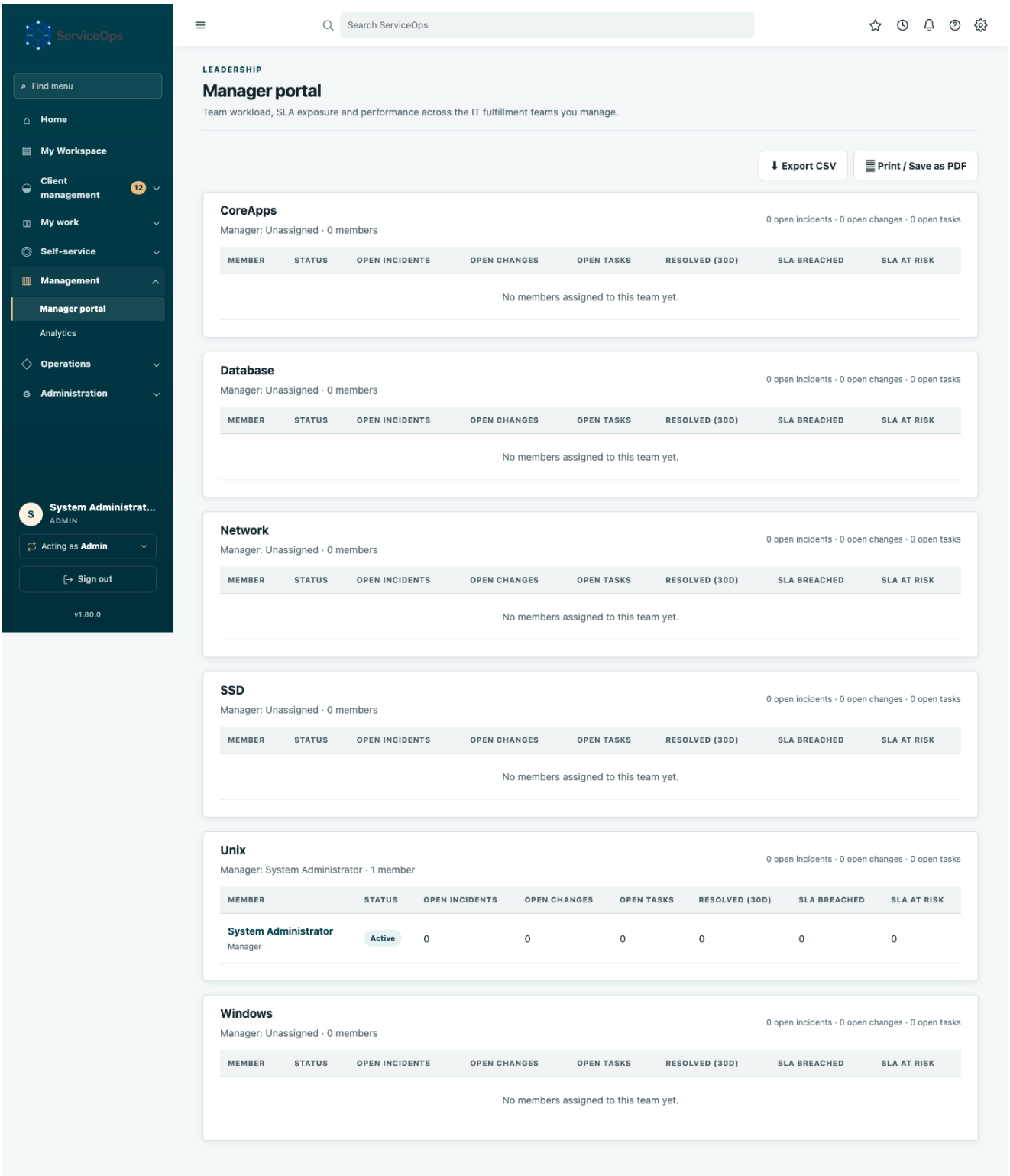
Org chart



A vertically indented org chart showing manager-to-report reporting lines, with each person's name, title, department and a role badge.

The org chart is generated automatically from each user's configured manager field — there's no separate org-chart data entry, so it's always consistent with **Users & roles**. Nodes with reports show a collapse toggle so a large organization can be explored level by level instead of one long scroll.

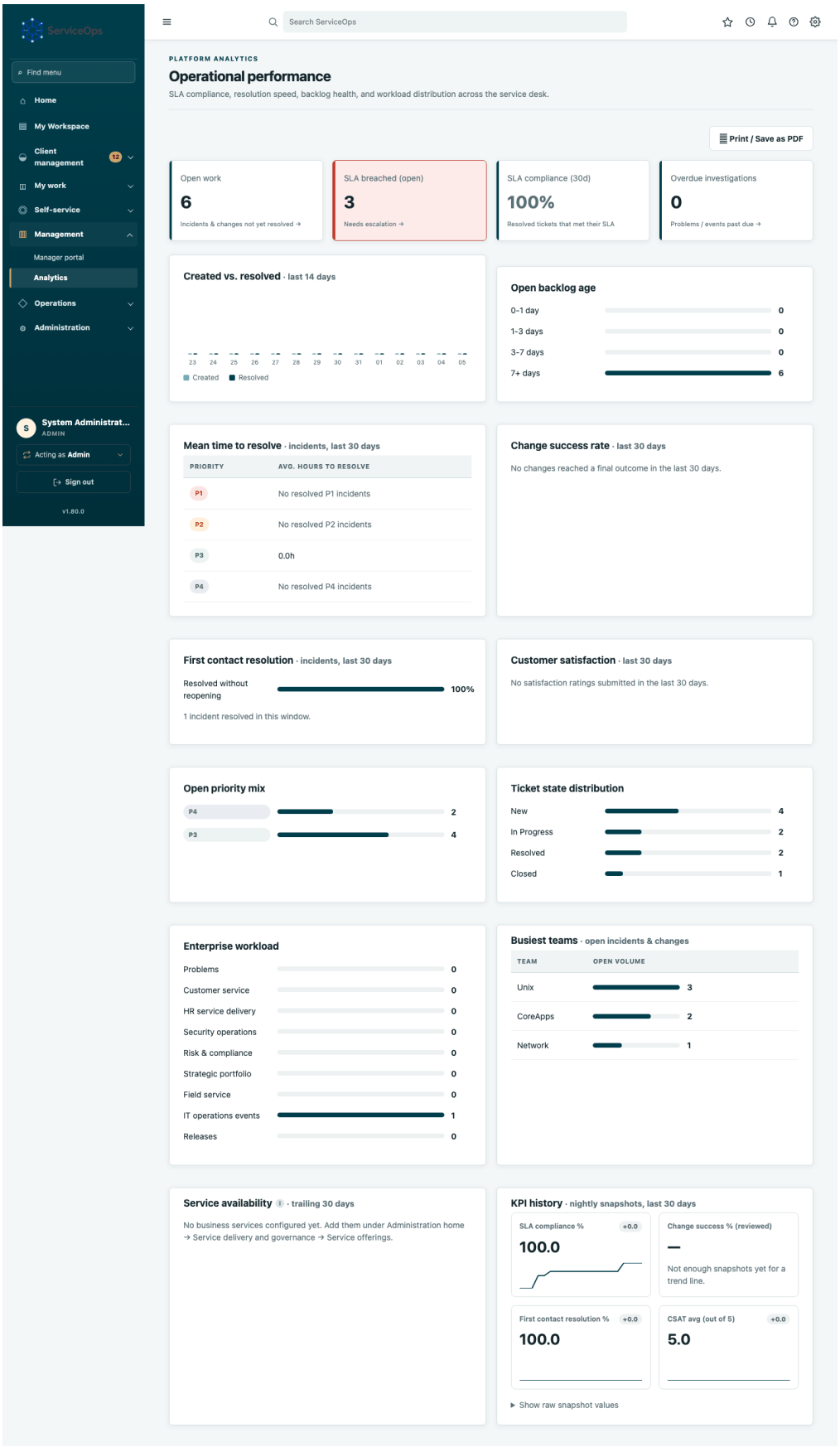
Manager portal



The manager portal: one panel per team showing team-level open incident/change/task counts and SLA-breach totals, followed by a per-member table with status, workload and SLA columns.

This is the primary reporting surface for a team manager. Each team you manage gets its own panel; inside it, every team member has a row showing their active/inactive status, how many incidents/changes/tasks are currently open against them, how many tickets they've resolved in the last 30 days, and their personal SLA breached/at-risk counts. The **Export CSV** and **Print / Save as PDF** buttons at the top let this exact view be pulled into a spreadsheet or a printed report for a status meeting without re-deriving the numbers by hand.

Analytics

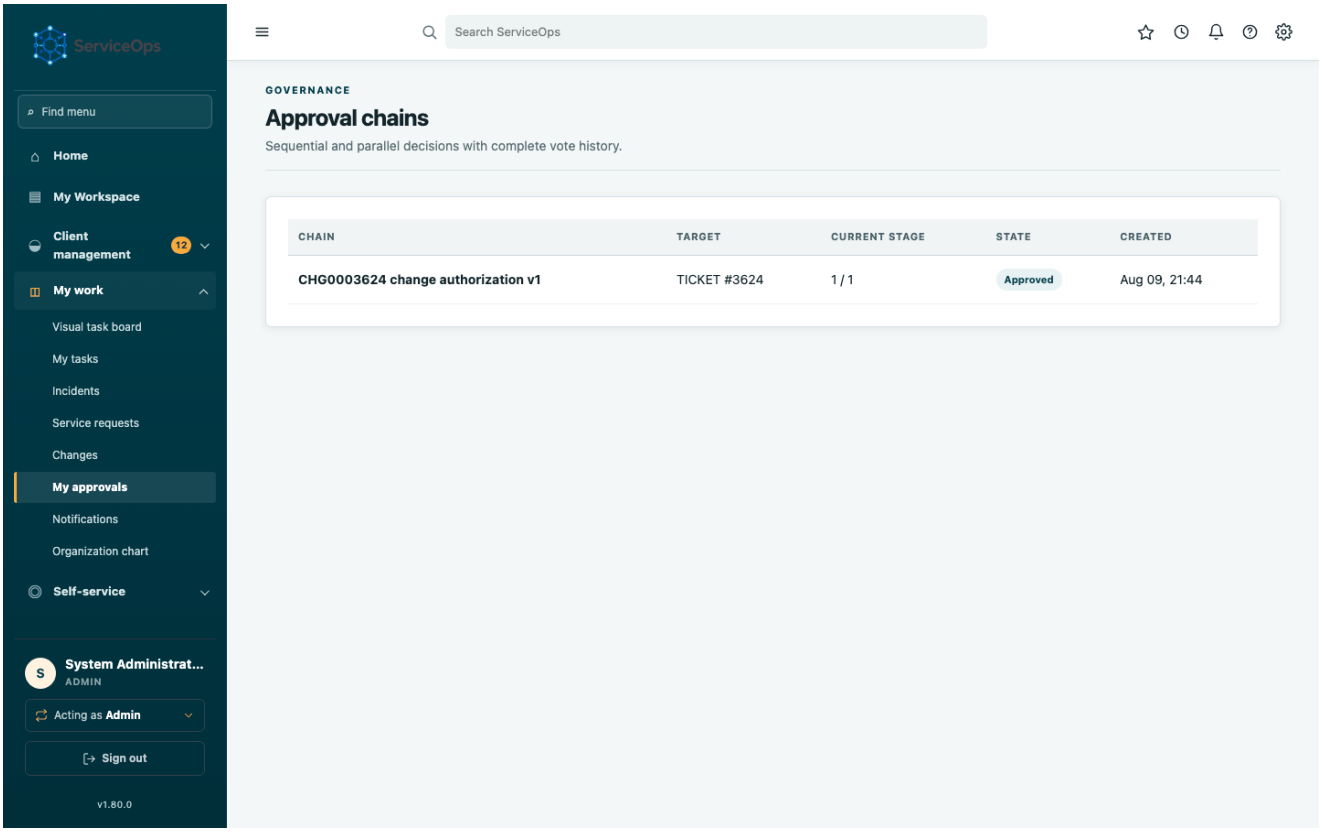


The analytics page: SLA compliance and breach/at-risk stat tiles, a 14-day created-vs-resolved trend chart, backlog aging buckets, mean-time-to-resolve by priority, change success rate, priority/state distribution bars, and a busiest-teams ranking.

Analytics is the tenant-wide operational-performance view, distinct from the manager portal's per-team/per-person focus. The created-vs-resolved trend chart is the fastest way to see whether the backlog is growing or shrinking week

over week; backlog aging shows how long currently-open tickets have been sitting, which is often a better early-warning signal than raw open counts; and mean-time-to-resolve by priority is the number most worth tracking over time as a service-desk health metric.

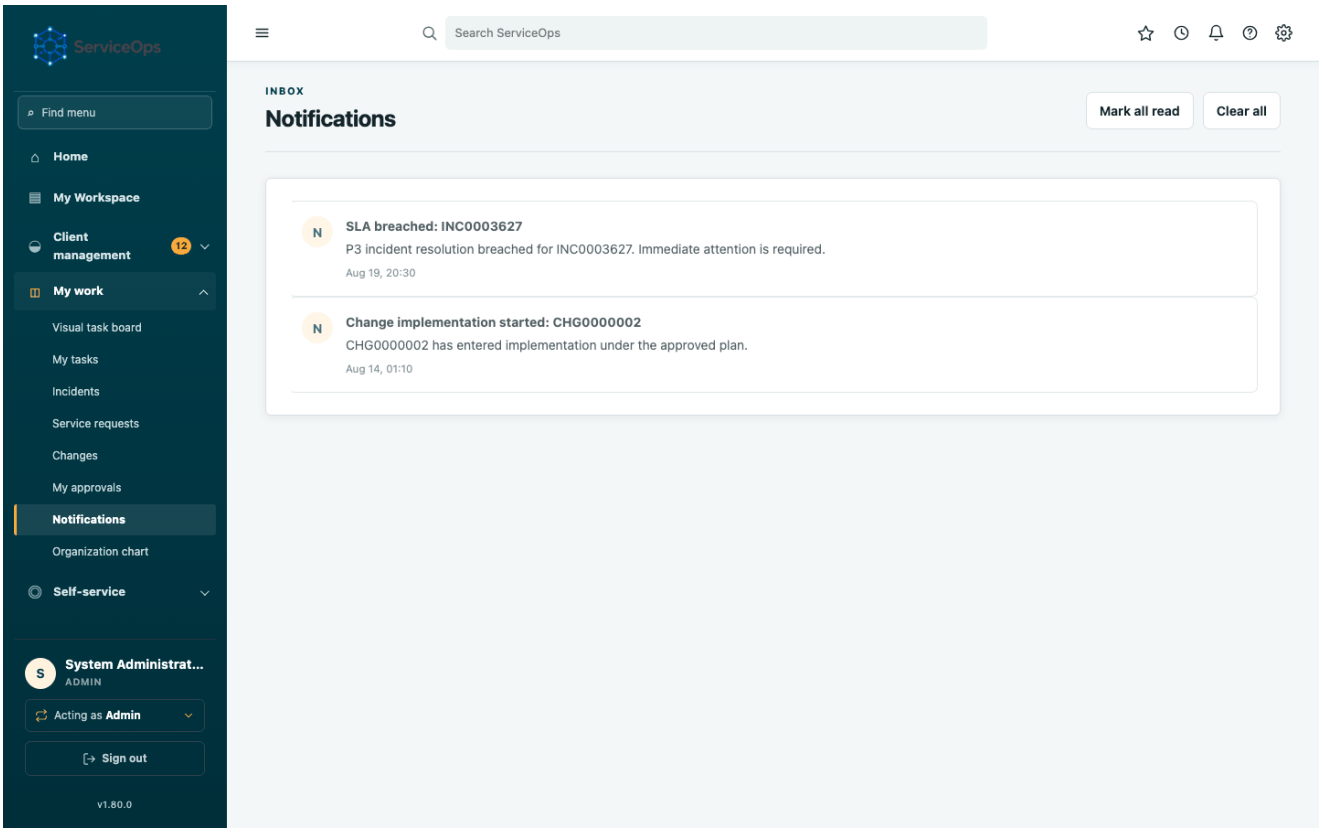
My approvals



The My approvals page: pending approval chains grouped by target record, each showing its gates, the votes already cast, and inline approve/reject controls for the ones waiting on the signed-in user.

Anyone named as an approver — a team manager, a CCB member — sees their pending decisions here rather than having to remember to check every change or RITM individually. The sidebar's "My approvals" link carries the same amber badge treatment as "My tasks": if there's nothing waiting on you, there is no badge; if there is, the count is right there before you open the page.

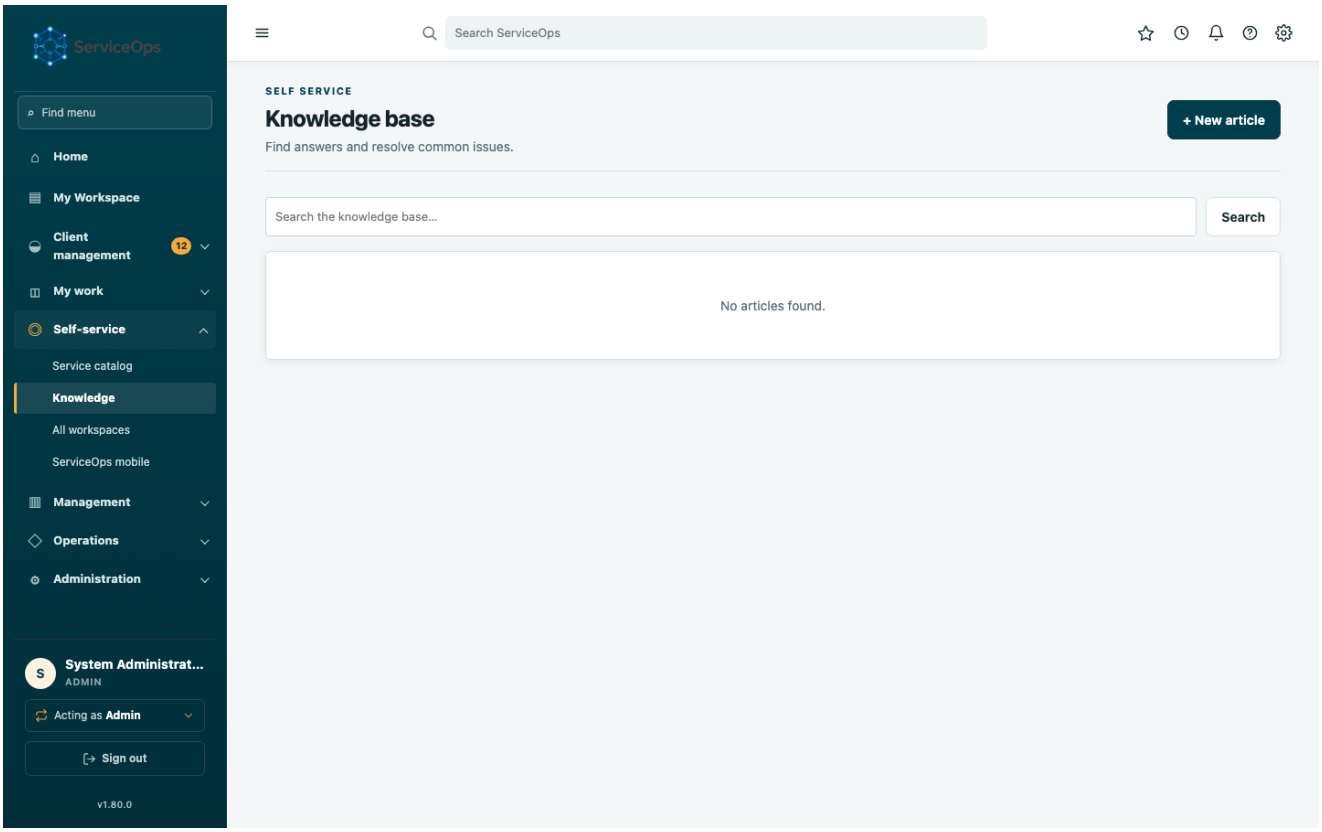
Notifications



The notifications inbox: a list of system notifications with unread items visually distinguished by a colored left border and accent dot.

Notifications are generated for the events you'd actually want to react to — an SLA breach on your ticket, a new approval request, a comment on something you're following. Unread items stay visually distinct until you click through to the record they refer to; opening the notifications page itself no longer silently marks everything read the moment you look at it.

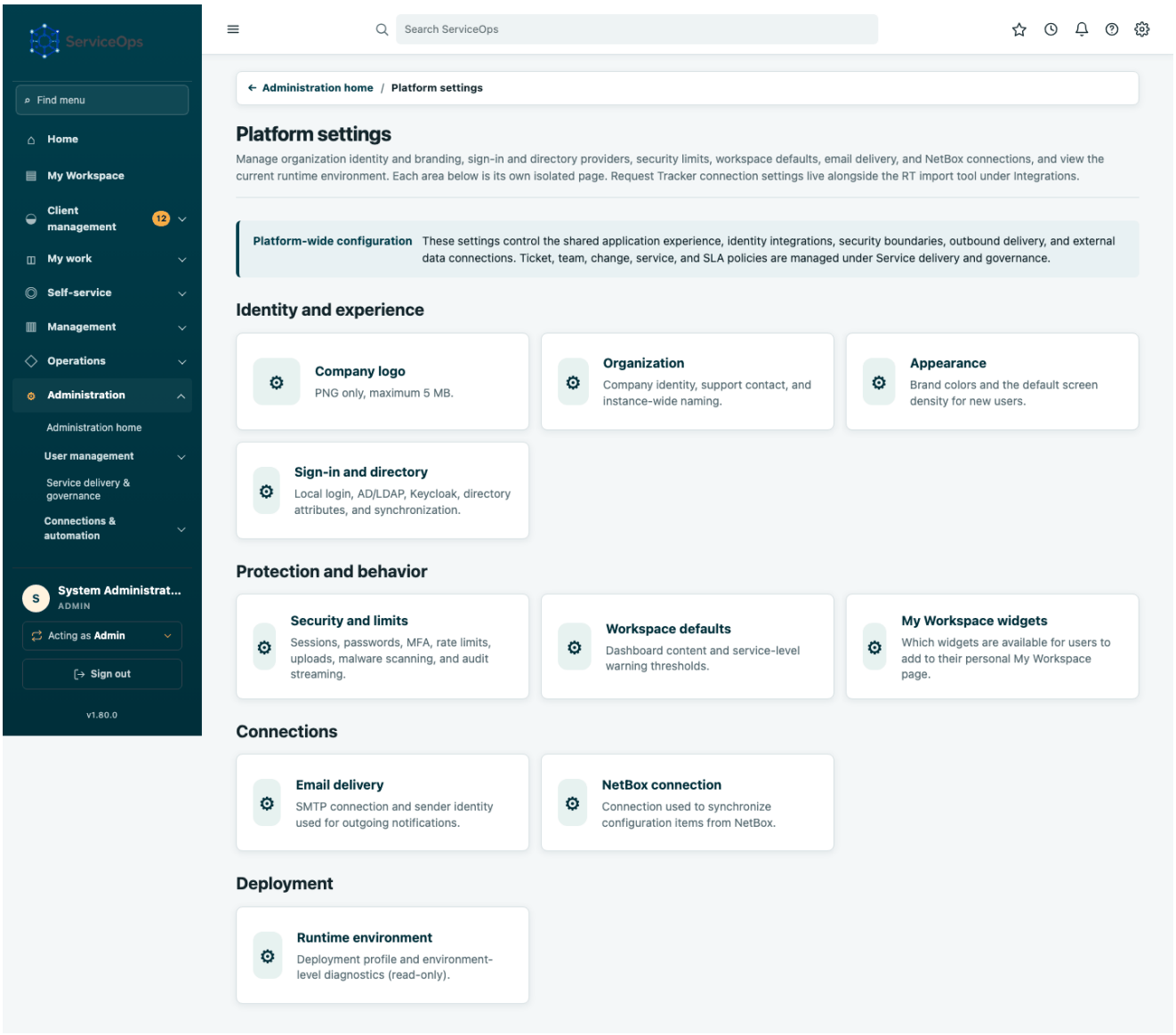
Knowledge base



The knowledge base: a card grid of published articles with title, summary and category.

Search knowledge before opening a duplicate incident or RITM — a workaround or a known error entry here can resolve a user's problem immediately without creating new operational work at all. Problems (PRB records) link back to the knowledge article that documents their permanent fix once root cause is found, closing the loop between "we investigated this once" and "here's how to solve it fast next time."

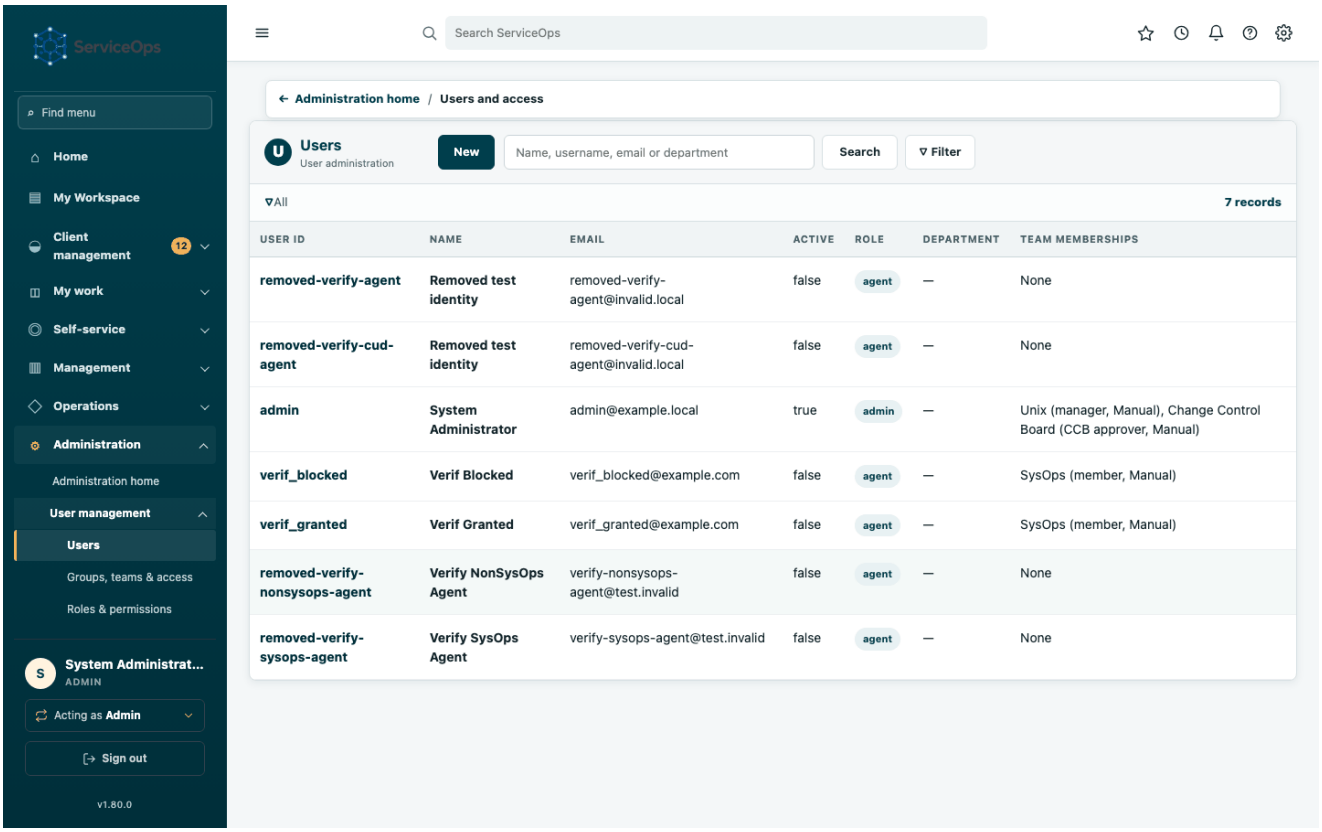
Administration home → Platform settings



The Platform settings page: tenant-wide configuration divided into identity and experience, protection and behavior, connections, and deployment groups; each field is marked Live or Restart required.

Every tenant-wide behavior this manual describes as configurable — branding, security limits, dashboard panel visibility, SLA warning windows, the new REST API rate limit, LDAP/Keycloak connection details — is set from this one page. Each field's badge tells you whether a change takes effect immediately (**Live**) or needs a full application restart/rollout to propagate to every running instance (**Restart required**) — this distinction matters most in a multi-replica Kubernetes deployment, where a Live setting is read fresh from the database by every request but a Restart-required identity-provider change must reach every pod consistently.

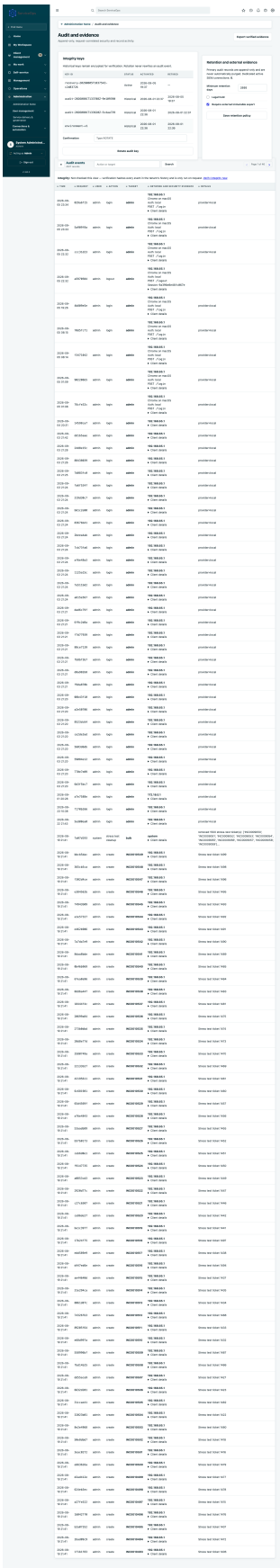
Users & roles



The Users & roles administration page: a searchable table of every user with username, name, role, department, active state and team membership, and a link into each user's editable record.

Administrators manage the full user lifecycle here: role assignment, active/ inactive state, department, manager (which drives the org chart), and team membership. AD/LDAP-sourced accounts continue to reconcile their team membership automatically from directory group mappings on every login: an administrator editing team membership here only affects locally-managed accounts, or overrides that get reconciled again on the next directory sync.

Audit log

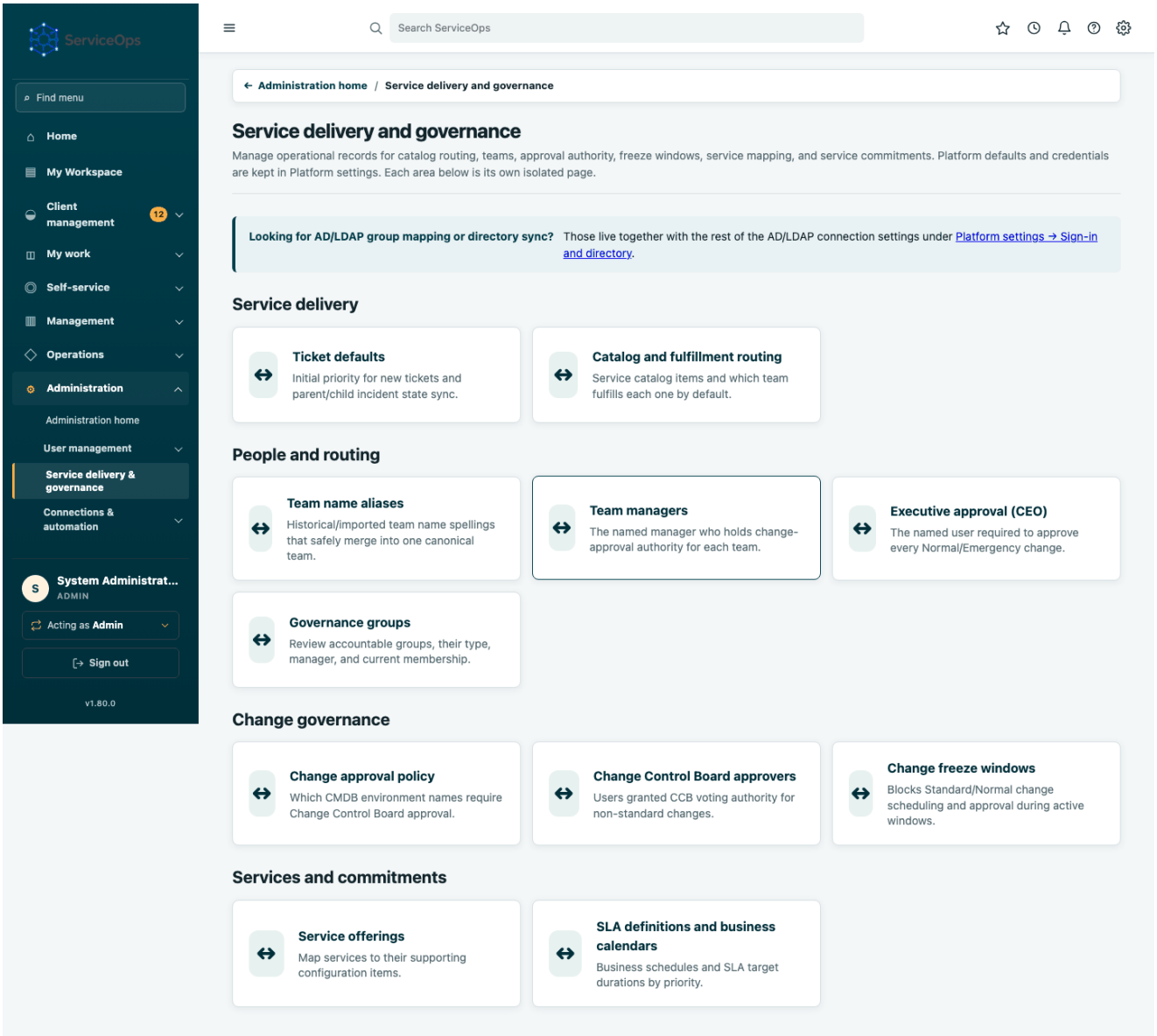


The audit log: a chronological, tamper-evident table of every recorded action across the tenant — actor, action, target and timestamp.

Every create, update, approval decision, and administrative action is recorded here with a cryptographic hash chain linking each event to the one before it, so the log itself can prove whether it has been tampered with. This is the

evidence trail an external auditor or incident responder would ask for first; see "Ticket history and change reapproval" earlier in this manual for how per-record history relates to this tenant-wide audit stream.

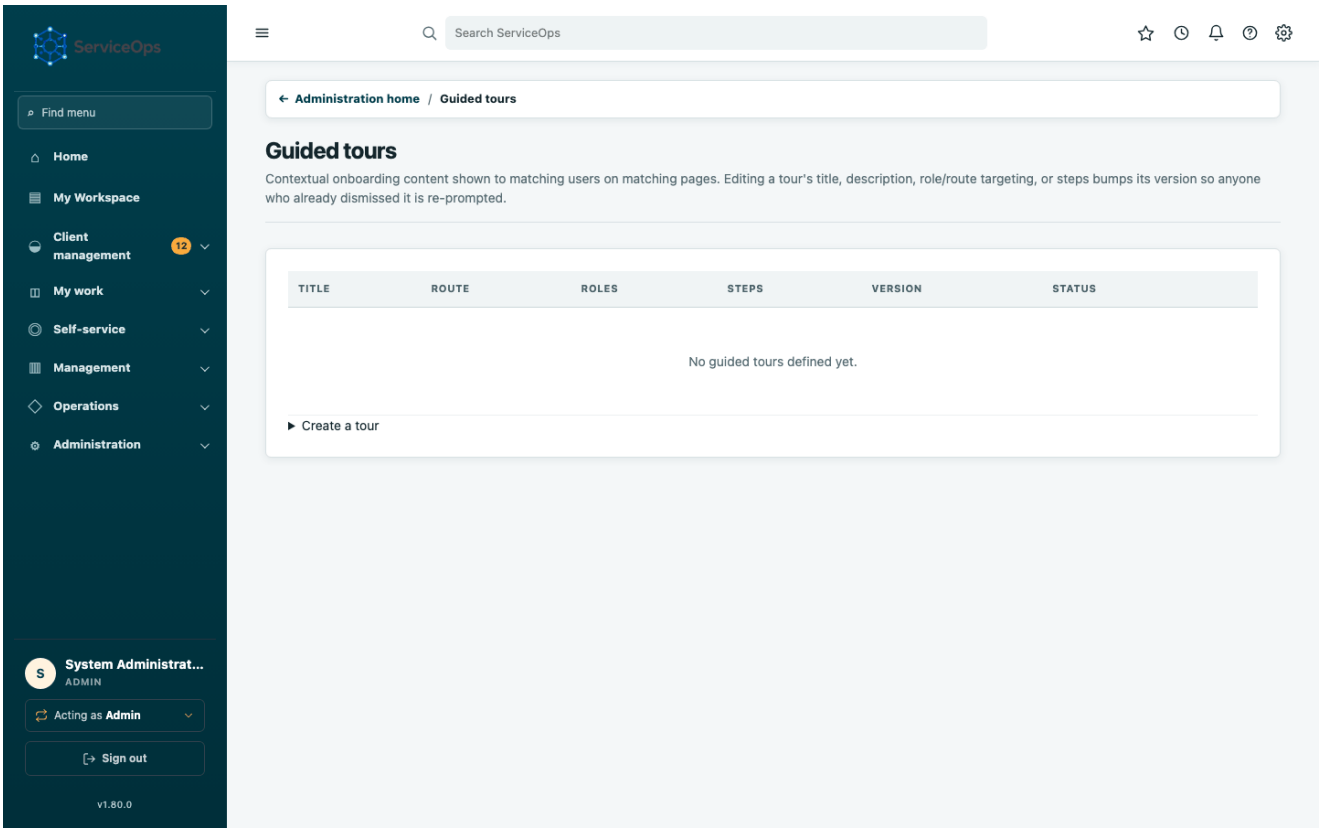
Administration home → Service delivery and governance



The Service delivery and governance page: operational records for catalog routing, people and teams, change governance, service mapping, calendars, and commitments.

This is where the process framework itself is configured, as opposed to day-to-day operational work: which team fulfills which catalog item by default, what an SLA's target duration and business calendar are, and who holds CCB approval authority. Changes made here govern all future tickets and requests; they do not retroactively alter SLA targets already attached to an in-flight ticket.

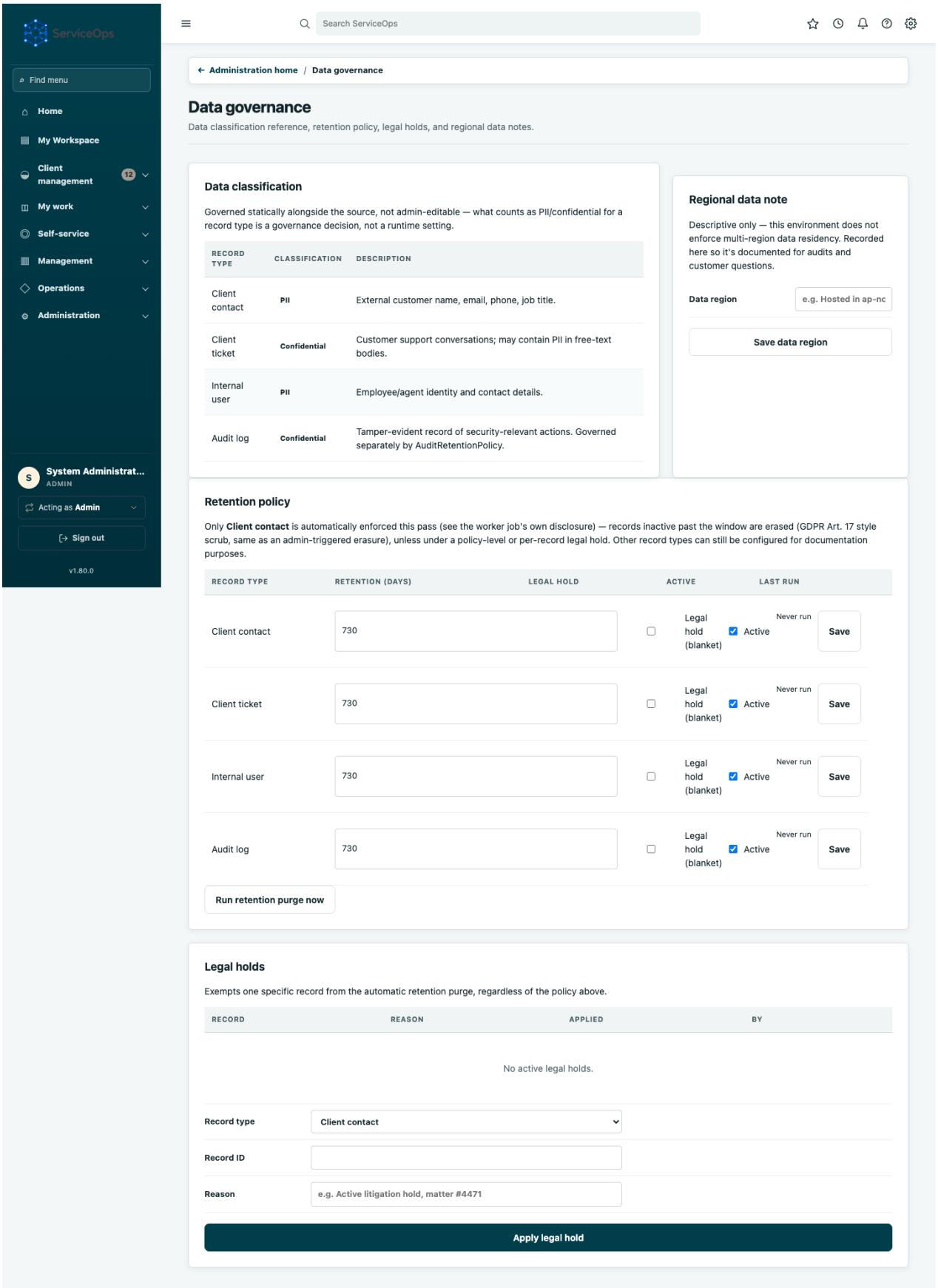
Guided tours administration



Guided tours administration: a list of admin-authored, versioned tours with ordered steps and role targeting, plus the tour-creation form.

Administrators author step-by-step guided tours here (`/admin/guided-tours`) — ordered steps, versioning, and role targeting so a tour can be scoped to just the roles it's relevant for. End users see active tours play back as a lightweight on-page overlay; each user's progress through a tour is recorded so it doesn't replay from the start on their next visit.

Data governance administration

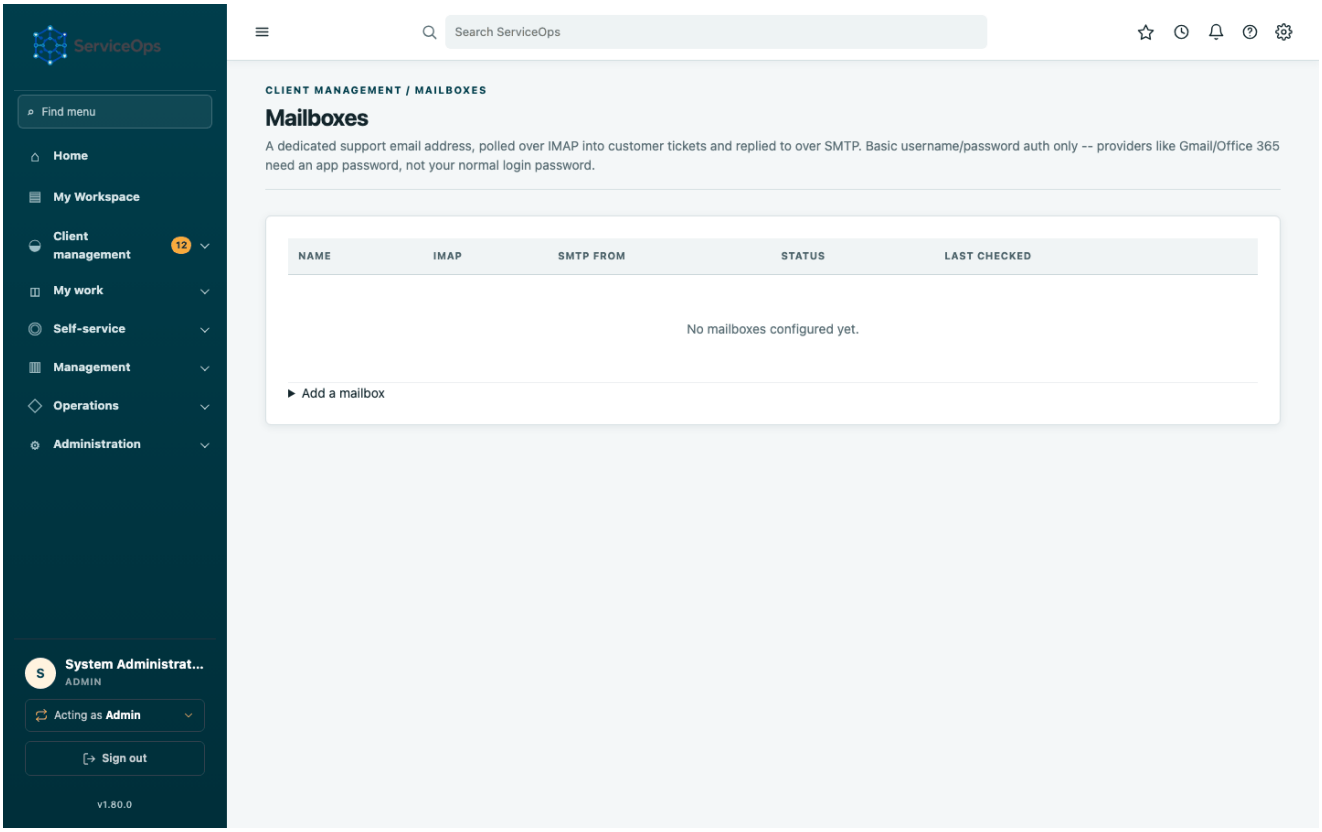


Data governance administration: retention policies, legal holds, and data classification, plus a queue for reviewing GDPR-style export/erasure requests.

The data governance page (`/admin/data-governance`) is where retention policies and legal holds are defined, and where classification-driven GDPR Art. 17/20-style erasure and export requests for client contacts are reviewed and

actioned. A legal hold on a record blocks its scheduled retention-driven deletion regardless of policy, until explicitly lifted.

Client support mailbox administration



Client support mailbox administration: a list of configured IMAP/SMTP-polled inboxes with active/inactive toggles, and the mailbox-creation form.

The Client Management support mailbox page (`/client-management/mailboxes`) configures a real IMAP/SMTP inbox that automatically becomes Client tickets: inbound mail is polled, threaded onto existing tickets by Message-ID/subject token, and creates a new ticket (with contact/organization auto-matching) when nothing matches; agent replies go back out through the same mailbox's SMTP, correctly threaded. Only basic-auth IMAP/SMTP is supported — providers like Gmail/Microsoft 365 that require OAuth or an app password need one configured here rather than the account's normal password.

3B. Common task walkthroughs

Report and resolve an incident

1. Click **+ Report an incident** on the dashboard, or **New** from the

Incidents list.

2. Provide a concise title, observable symptoms, category/subcategory and

business impact/urgency; ServiceOps calculates priority from impact and urgency automatically.

3. Submit — the incident is created in state `New` and routed to its owning

fulfillment team.

4. An agent from that team opens the record (see the Incident detail

screenshot above), sets it `In Progress`, and works the issue: comments, attachments, checklist items and CI links are all recorded in the Event history as they happen.

5. Once service is restored, the agent sets state to `Resolved`; a manager or

the same agent later moves it to `Closed` once confirmed with the caller.

Submit a Normal change through approval

1. From the Changes list, click **New**, select change type `Normal`, and

complete the required implementation plan, test plan, backout plan, planned window, primary configuration item and owning team; use **Add another configuration item** if the change touches more than one CI.

2. On submission ServiceOps runs conflict detection against other in-flight

changes touching the same CI or a related one, and creates a two-stage approval chain: the owning team's manager first, then the CCB.

3. The change sits in `Awaiting Approval` until both gates are satisfied —

visible on **My approvals** (see the amber sidebar badge) for anyone named as an approver.

4. Once approved, the change moves to `Approved` and its change tasks (CTASK)

can begin; a required CTASK still open blocks the change from completing.

5. Editing a material field (plan, risk, affected CI, schedule) after

approval automatically supersedes the existing chain and starts a new one — an old approval is never silently applied to a changed plan.

Order an item from the service catalog

1. Open the **Service catalog** page, pick an item, add any requested

details, and click **Request**.

2. ServiceOps creates a REQ container and one RITM for the item; if the item

requires approval, an approval chain starts (the requested-for employee's recorded line manager, then Service Desk fulfillment authorization) before any fulfillment work begins. If either stage has no valid approver, the request fails closed before records are created.

3. Track status from **Requests & RITMs** — open the RITM itself to see its

own lifecycle stepper and the SCTASKs doing the actual fulfillment work.

4. The RITM reaches `Closed Complete` once every SCTASK finishes; the parent

REQ reaches its own terminal state once every RITM in it does.

Review team performance as a manager

1. Open **Management → Manager portal** from the sidebar.

2. Each team you manage has its own panel with team-level open-work and

SLA-breach totals, and a per-member table below it.

3. Use the per-member columns (open incidents/changes/tasks, 30-day resolved

count, SLA breached/at-risk) to spot who's overloaded or who has SLA risk building up before it becomes a breach.

4. Use **Export CSV** to pull the exact table into a spreadsheet, or

Print / Save as PDF to produce a printable report for a status meeting — both reflect exactly what's on screen.

4. Deployment decision

Environment	Application	Database	Upload storage
Single server	Docker Compose	Bundled PostgreSQL	Docker volume
Single production server	Docker Compose behind HTTPS proxy	Bundled or external PostgreSQL	Docker volume with backups
Enterprise production	Kubernetes/Helm, 3+ replicas	Managed HA PostgreSQL	RWX CSI volume or object-storage extension

Production Kubernetes must use an externally operated, highly available PostgreSQL service.

5. Docker installation

Run `./serviceops install web`, open `http://127.0.0.1:8090`, configure the profile, database, identity providers, and listener, validate all checks, confirm, and deploy. Keep the terminal open until the browser reports a healthy deployment.

Use `./serviceops status`, `health`, `logs`, `backup`, `restore`, and `doctor` for lifecycle operations. Put Caddy, NGINX, or an enterprise load balancer in front of the loopback-bound application and terminate TLS there.

6. Kubernetes prerequisites

- Kubernetes 1.27 or newer with at least three worker nodes for HA.
- Helm 3, kubectl, a default-deny-capable CNI, and a CSI storage provider.
- A private image registry and immutable ServiceOps image tag or digest.
- External HA PostgreSQL with TLS, automated backups, and point-in-time recovery.
- RWX upload storage when application replicas exceed one.
- An ingress controller, DNS, and automated TLS certificate management.
- Metrics Server for HPA; cluster monitoring and log aggregation.
- A secrets controller or external vault for steady-state secret rotation.

7. Kubernetes installation

1. Build, scan, sign, and push an immutable image.
2. Copy `deploy/kubernetes/values-production.example.yaml` to

`deploy/kubernetes/values-production.yaml`.

3. Set registry, immutable tag, ingress, storage class, replicas, identity

endpoints, and role mappings.

4. Run `./serviceops install kubernetes --preflight`.
5. Run `./serviceops install kubernetes`.
6. Confirm rollout, Helm test, ingress TLS, login, record creation, upload,

approval, notification, and audit behavior.

The installer labels the namespace for Restricted Pod Security and creates separate operator-managed runtime and bootstrap Secrets from mode-`0600` temporary files on the first install. It preserves both on upgrades and fails closed if only one exists; the chart itself never renders plaintext credentials into Helm release state. Escrow the runtime Secret

before deployment because rotating its encryption, audit-integrity, or token-pepper keys can break data or verification continuity. The installer then uses `helm upgrade --install --atomic --wait`, waits for rollout, and runs the packaged `/ready` test.

For an upgrade, provide both the release tag and verified digest; changing only the tag cannot update a digest-governed workload:

```
SERVICEOPS_VALUES=deploy/kubernetes/values-production.yaml \
./tools/safe_update_k8s.sh <stable-tag> sha256:<verified-64-character-digest>
```

Never use `kubectl set image` for ServiceOps. It creates configuration drift, bypasses the migration hook, and disagrees with the Helm release record.

8. Kubernetes chart controls

- Non-root UID/GID, RuntimeDefault seccomp, all capabilities dropped.
- Read-only root filesystem and disabled service-account-token mounts.
- Startup, readiness, and liveness probes with distinct semantics.
- Rolling update with zero unavailable replicas.
- Topology spreading across zones and hosts.
- PodDisruptionBudget and optional HPA.
- Resource requests and limits.
- NetworkPolicy for ingress and required egress ports; production ingress

requires an explicit trusted namespace selector.

- Persistent upload claim and optional ingress/TLS.
- JSON schema validation and a Helm test hook.
- A pre-install/pre-upgrade migration Job that waits for PostgreSQL.
- Web/worker init gates that wait for connectivity and the exact Alembic head.
- A bounded, NetworkPolicy-isolated Helm readiness test using a digest-pinned

helper image; it may reach only DNS and ServiceOps web pods, and its completed pod is retained long enough for `helm test --logs` evidence.

Tune topology keys to the labels present in the target cluster. A PDB protects only against voluntary disruptions; it does not protect against node, zone, or application failure.

9. Identity configuration

Local administrator

Keep local authentication enabled for one vaulted break-glass administrator. Test it quarterly under an approved access procedure. Rotate after every use.

AD/LDAP

Use LDAPS or StartTLS, certificate validation, a least-privilege read-only bind identity, a narrow base DN, an escaped username filter, and explicit group-role mappings. Validate bind, search, user bind, disabled-user behavior, group mapping, certificate expiry, and directory outage behavior.

The self-service profile shows the reporting line, direct reports, team responsibilities and a bounded directory snapshot. The default attribute map supports display/given/family name, title, department, division, employee ID and

type, office, telephone/mobile, mail, UPN, manager, team, group memberships, account control/expiry and selected directory/logon timestamps, country, object GUID, uid/gid, Unix home and login shell. This is an allowlist: password-related material, certificates, SIDs and raw security descriptors from an unrestricted LDAP search are never persisted or rendered. Review the map against the organization's schema and data-minimization policy before enabling additional attributes.

Keycloak

Use a confidential OIDC client, authorization-code flow, exact HTTPS redirect URI, short-lived tokens, approved realm-role claims, client-secret rotation, and restrictive redirect/web-origin settings. Validate login, logout, expired session, revoked user, missing email, role changes, and provider outage.

In addition to name/email/role claims, Keycloak/OIDC login can populate the same profile fields LDAP sync does — title, department, division, employee ID, employee type, business phone, mobile phone, and location — from OIDC userinfo claims, via the admin-configurable `KEYCLOAK_ATTR_MAP` setting (Administration home → Platform settings → Sign-in and directory), a `{ServiceOps field: claim name}` JSON map mirroring `LDAP_ATTR_MAP`'s shape. Populate it with a realm's actual custom protocol-mapper claim names (e.g. `department`, `employee_id`, `phone_number`); an empty map skips profile mapping and only name/email/roles are applied, as before. As with LDAP sync, a sparser claim set on a later login never nulls out a profile value a previous login already set.

LDAP directory synchronization

ServiceOps deliberately does not provide a one-click full-directory import. Successful LDAP authentication provisions the user just in time. Scheduled, tenant-scoped reconciliation enriches existing profiles and creates only a missing manager needed to preserve an approval chain. It enriches profile fields (`title`, `department`, `division`, `employee_id`, `employee_type`, `business_phone`, `mobile_phone`, `location`), bounded directory details shown on the profile, the manager reporting chain (`User.manager_id`), account enabled/disabled state, and AD-group-driven team membership. The safe detail snapshot also normalizes the directory OU and DNS domain, website, Unix username/UID/GID/home/shell, NIS domain, account type, and useful security timestamps such as the last bad-password attempt and lockout time. Group DNs are reduced to friendly names and summarized by their OU purpose; team-like names are shown as administrator-review suggestions but are never silently assigned. Both the self-service profile and administrator user record connect the identity to assigned assets and personally owned CIs, and the GDPR data export includes those operational relationships. This is implemented in `serviceops_core/ldap_sync.py::sync_directory`. It has no bulk-provisioning mode or manual HTTP trigger.

The `worker` container's existing in-process polling loop (`tools/outbox_worker.py`, the same loop that already processes SLA breaches, workflow schedules/jobs, and the outbox) also calls `app.process_ldap_sync_schedule()` on every pass. For each **active** tenant with both `LDAP_ENABLED` and `LDAP_SYNC_ENABLED` set, it checks a per-tenant `ldap_sync_state` table row and runs `sync_directory(tenant.id)` once the configured interval has elapsed since that tenant's last run. A failure syncing one tenant is logged (no secrets) and never blocks or crashes the sync of other tenants or the rest of the worker loop. There is no default or fallback tenant — iteration is always by explicit, individually-flagged tenant, consistent with the platform's fail-closed tenant policy. Runs use the configured LDAP scope, RFC 2696 pages, an entry safety ceiling, and at least a 15-minute cadence. Failures cool the tenant down to four times the configured interval. Runs refresh linked users and reporting managers but never enable full-directory account creation.

Relevant settings (Administration home → Platform settings → Sign-in and directory):

Setting	Purpose
`LDAP_ATTR_MAP`	JSON map from ServiceOps profile/manager/email/username/directory-detail fields to the directory's actual attribute names (e.g. AD `employeeID` vs. an OpenLDAP equivalent). Defaults cover contact/location, team, website, Unix/NIS identity, account type, account-security timestamps and `account_control`→`userAccountControl`.

Setting	Purpose
`KEYCLOAK_ATTR_MAP`	The same shape of JSON map, applied to Keycloak/OIDC userinfo claims instead of LDAP attributes (see **Keycloak** above). Independent of `LDAP_ATTR_MAP` — a tenant can run either or both providers with their own mappings.
`LDAP_SYNC_ENABLED`	Enables scheduled reconciliation for this tenant. Default off.
`LDAP_SYNC_INTERVAL_MINUTES`	Minimum minutes between runs per tenant (15–10080). Default 60.
`LDAP_SYNC_PAGE_SIZE`	RFC 2696 page size (25–500). Default 100. Keep it below the directory server's configured maximum.
`LDAP_SYNC_MAX_ENTRIES`	Hard per-run safety ceiling (100–50000). Default 5000; reaching it warns the administrator to narrow `LDAP_USER_FILTER`.
`LDAP_AUTO_CREATE_TEAMS`	Conservatively creates a tenant-scoped team only from the configured single-valued team attribute. Default off. It never turns every `memberOf` group into a ServiceOps team.
`LDAP_SYNC_ACCOUNT_STATUS`	Deactivates a linked ServiceOps account when AD marks it disabled, revoking its sessions. Default on. Re-enablement remains an explicit administrative decision.
`CLIENT_HOSTNAME_LOOKUP`	Optionally records a forward-confirmed reverse-DNS name for a login source IP. Default off. The result is informational and never an identity or authorization input.

Operational notes:

• Directory attributes that are absent or empty are never used to null out

existing ServiceOps values (sparse directory entries do not erase data).

• A manager DN outside the search filter/base DN is left unresolved rather

than failing the whole sync.

• All directory attribute names are admin-configurable; nothing about a

specific company's schema is hard-coded.

• Reporting links are applied before manager grants are recalculated, avoiding

order-dependent role flapping. A user with an active direct report or managed team receives manager capability automatically.

• If automatic team creation is enabled, manager inference is fail-safe: it is

applied only to an unowned team when every active member has the same active, same-tenant line manager. Explicit managers are never replaced.

• This does not populate org-chart/manager data for interactively-created

local accounts that have never authenticated via LDAP.

Manager absence approval coverage

A people or team manager may open **My profile** and record a start, end and reason for an absence. Coverage is permitted only upward to that manager's own active, same-tenant line manager and for at most 90 days. Existing requested votes remain accountable to the original manager until the backup acts; at decision time ServiceOps stores the backup as the actor and the original manager as `delegated_from`. The backup intentionally receives no initial

request notification, and web/mobile decisions enforce the same state, tenant, authorization and change-freeze policy. Cancellation and decisions are audited. Configure reporting lines before relying on this control and test the organization's real absence/escalation scenario during go-live validation.

10. Post-deployment system settings

Administrators can open **Administration home → Platform settings** after deployment to change the platform name, company name, PNG logo, primary and accent colors, support identity, display defaults, LDAP, Keycloak, encrypted provider secrets, security limits, workflow defaults, and notification identity.

Each field is marked either **Live** or **Restart required**. Live settings are read from PostgreSQL by every request. Restart-required identity-client changes must be followed by a complete Compose restart or Kubernetes rollout so every application instance receives the same client configuration.

Database topology, replicas, storage, ingress, and TLS remain controlled by Compose or Helm and are displayed read-only. This prevents one pod from silently diverging from the declared infrastructure.

Sensitive values are encrypted before storage and are never returned to the browser. Set a durable `SETTINGS_ENCRYPTION_KEY` during installation; changing or losing that key makes existing encrypted settings unreadable.

Priority and SLA policy

Ticket priority is calculated from impact and urgency using the validated, Git-backed `config/priority_matrix.json`. Agents cannot silently override the result. A manager or administrator may select a different priority only while supplying an auditable reason of at least ten characters.

Administrators manage business calendars and SLA definitions under **Administration home → Service delivery and governance**. Calendars use IANA timezone names, explicit business weekdays and opening hours, plus named excluded holiday dates. An SLA without a calendar remains a 24x7 wall-clock commitment. A calendar change applies to future SLA attachments; it does not silently rewrite targets already in flight.

The worker detects newly breached commitments, writes one breach event to the SLA evidence stream, records the breach in the ticket history, and notifies the owning-team manager and assigned engineer through the durable outbox. Monitor the worker continuously; a stopped worker delays escalation but does not lose the stored target or breach state.

Declarative workflow operations

Workflow source is controlled in `config/workflows.json`. On startup ServiceOps validates the package and publishes a new immutable PostgreSQL runtime version only when the canonical specification changes. Administrators can inspect the deployed hash and versions, redeploy the packaged source, and run a mutation-free simulation under **Administration home → Automation rules**.

The supported foundation accepts `ticket.state_entry` events, equality, inequality, membership and empty-value conditions over explicitly allowed ticket fields, plus three actions: add ticket history, notify the requester, and notify the owning-team manager. Arbitrary administrator scripts and unsupported fields/actions fail package validation.

State transitions enqueue a unique correlated job in the same transaction as the operational change. The worker claims jobs using PostgreSQL coordination, records the exact published version and masked structured input/output, and commits actions atomically. Failures use bounded exponential retry and enter `Dead` after five attempts. Monitor both workflow jobs and execution evidence from the administration page; investigate dead jobs before replaying them.

Release 1.13.0 extends the runtime with durable waits, per-action execution evidence, manual/API/SLA-breach triggers, per-workflow rate limits, controlled dead-job replay, and safe terminal compensation. A wait commits its cursor and resume timestamp to PostgreSQL; after restart the worker continues at the next action and does not repeat completed steps.

API workflow triggers require the explicit `workflows:execute` scope, the acting user's operational permission for the ticket, and an idempotency key. Rate-limited jobs are delayed without consuming a failure attempt. Failed jobs retry five times; compensation runs only after the terminal failure and is currently restricted to an auditable ticket-history action because delivered notifications cannot truthfully be undone.

Release 1.13.0 does not yet claim scheduled recurrence, reusable subflows, general rollback, broad record mutation, concurrency quotas, or complete multi-environment configuration promotion. Those remain governed backlog work.

Release 1.14.0 adds reusable subflows to workflow package schema v2. Subflow references are validated before deployment, unknown dependencies fail closed, and dependency cycles are rejected. Valid subflows are expanded into the immutable published workflow specification, so runtime execution never depends on mutable external fragments.

Administrators configure recurring ticket schedules under **Administration home → Automation rules**. Each schedule is tenant-scoped, targets one ticket, uses a bounded minute interval, and can be enabled or disabled without deleting its evidence. Concurrent workers claim due schedules with PostgreSQL row locking. The event and next-run advancement commit together. If the system was offline for several intervals, it emits one event and advances directly to the next future interval instead of creating a catch-up storm.

Calendar expressions, blackout windows, package dependencies, per-tenant concurrency quotas and production-scale scheduler failure testing remain governed backlog work.

Bootstrap credential retirement

ServiceOps reads `ADMIN_PASSWORD_FILE` before the legacy `ADMIN_PASSWORD` environment value. Kubernetes separates the bootstrap credential from runtime secrets and never injects it into workers. Compose workers likewise receive no bootstrap administrator credential.

After the first successful installation:

1. Sign in as the local administrator.
2. Use **Administration → Change password** to rotate the initial credential

into a unique vault-managed password. This increments the account authentication version and invalidates every other browser session.

3. Store the new credential in the organizational emergency-access vault.
4. Run `./serviceops retire-bootstrap-secret`, verify the active administrator,

type the explicit confirmation, and wait for the recreated containers to pass health checks.

5. Run `./serviceops doctor`; the worker-secret isolation check must pass.

The retirement operation clears only bootstrap injection from `.env`; it does not erase or reset the administrator's salted password hash in PostgreSQL.

Release evidence

Tagged releases run `.github/workflows/supply-chain.yml`. Every referenced GitHub Action is pinned to a full commit SHA. The gate runs the complete test suite, builds with maximum provenance, blocks fixable HIGH and CRITICAL operating-system or library vulnerabilities, generates a CycloneDX image SBOM, and publishes only after successful validation. The resulting digest is signed keylessly with GitHub OIDC and receives GitHub SLSA build-provenance and SBOM attestations. Registry digest inspection and `gh attestation verify` must both succeed.

The opt-in protected production workflow `.github/workflows/deploy-kubernetes.yml` runs after a successful governed release when `KUBERNETES_DEPLOY_ENABLED=true`. It consumes protected `KUBE_CONFIG_B64` and `KUBERNETES_VALUES_B64` environment secrets, verifies the released image provenance and digest, lints/renders the chart, performs `helm upgrade --install --atomic --wait`, verifies both Deployments, and runs the packaged readiness test. GitHub production-environment reviewers remain a separate human authorization boundary.

Kubernetes values must provide `image.digest`; application, worker and migration workloads use `repository@sha256:digest`, never a mutable tag. The installer requires `SERVICEOPS_GITHUB_ORGANIZATION`, installs the pinned Sigstore Policy Controller and GitHub trust policy, and labels the ServiceOps namespace for enforcement. Unsigned or untrusted matching organization images are rejected at admission. Run `python tools/verify_supply_chain.py` locally to detect release-policy drift.

Runtime Python and bundled PostgreSQL base images are pinned by digest and Python dependencies use exact versions. Generate the release record with:

```
python tools/release_evidence.py \
--version 1.26.3 \
--image serviceops-app:1.26.3 \
--output release-evidence/serviceops-1.26.3.json
```

The output records Git state, SHA-256 hashes for non-ignored source files, a combined manifest hash, image reference and CycloneDX component inventory. A dirty-tree marker is evidence, not approval. External vulnerability scanning, image signing, provenance publication, registry verification and admission enforcement remain required before organizational production approval.

11. Security operations

- Enforce TLS externally and enable HSTS only after HTTPS is proven.
- Store secrets in a vault; never commit `.env`, values secrets, or kubeconfig.
- Restrict namespace RBAC and database privileges.
- Scan source, dependencies, image, manifests, and exposed endpoints in CI.
- Sign images and enforce admission verification where supported.
- Forward application, ingress, Kubernetes audit, database, and identity logs.
- Alert on repeated login failures, admin changes, approval anomalies, SLA

breaches, crash loops, readiness loss, saturation, and storage pressure.

- Patch base images and dependencies under change control.

12. Backup and recovery

Back up PostgreSQL and uploads as one recovery set. Encrypt backups, store them outside the cluster, apply retention and immutability, and record restore test evidence. For managed PostgreSQL enable PITR. Snapshotting a running database volume alone is not a proven logical backup.

Quarterly recovery test:

1. Run `./serviceops rehearse-recovery` and `./serviceops rehearse-pitr`; retain

both evidence records.

2. Restore into an isolated environment.
3. Restore database to the selected recovery point.
4. Restore uploads from the matching recovery set.
5. Deploy the matching immutable application version.
6. Validate counts, attachments, identities, approvals, audit, and critical

workflows.

7. Record actual RPO/RTO and corrective actions.

The logical recovery command deliberately reports `pitr_proven=false`: `pg_dump` cannot be replayed with WAL. The separate PITR rehearsal uses `pg_basebackup`, continuous WAL archiving, and a named recovery target on disposable clusters, and reports `pitr_proven=true` only after boundary and ServiceOps integrity checks pass.

13. Upgrades and rollback

Run `./serviceops rehearse-upgrade` first. It creates a verified rollback set and validates the candidate image and migration against an isolated clone while the source remains healthy. Back up first. Review release notes and schema compatibility. Render and lint the Helm chart, deploy to staging, run workflow regression and load tests, then use `helm upgrade --install --atomic --wait`. Observe error rate, latency, readiness, database health, and queues. `--atomic` rolls Kubernetes resources back when the upgrade fails, but database schema/data rollback still requires a release-specific tested procedure.

14. Monitoring and SLOs

Monitor availability, request rate, latency percentiles, HTTP errors, worker saturation, pod restarts, readiness, CPU/memory throttling, database connection usage, query latency, replication/backup health, volume capacity, login errors, notification failures, and SLA breach rate. Define business-approved SLOs and page only on actionable symptoms.

15. Incident response

Declare severity and incident commander, preserve evidence, stabilize service, communicate on a fixed cadence, use tested rollback/failover, validate recovery, and create a blameless problem record with corrective actions. Never delete audit or operational evidence during response.

16. Production acceptance checklist

- ■ Immutable, scanned image from trusted registry
- ■ External HA PostgreSQL with TLS, PITR, and restore test
- ■ Three or more application replicas across failure domains
- ■ RWX persistent uploads and tested restore
- ■ TLS ingress, DNS, HSTS, security headers
- ■ Restricted Pod Security and least-privilege RBAC
- ■ Network policies validated with the actual CNI and endpoint topology
- ■ LDAP/Keycloak end-to-end tests and vaulted break-glass account
- ■ Monitoring, logs, alert routing, dashboards, and runbooks
- ■ Load, soak, failover, node-drain, rollback, and disaster-recovery tests
- ■ Security review, penetration test, and formal go-live approval